



LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 1 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

620 Old Peachtree Road NW, Suite 100, Suwanee, GA 30024 USA (770) 570-1800

Test Report Number: GZES240100172661

A representative sample of the product covered by this report has been evaluated and found to comply with the applicable requirements of:

- Inverters, Converters, Controllers and Interconnection System Equipment for Use With Distributed Energy Resources –UL 1741 (Third Edition, Dated September 28, 2021)
- Power conversion equipment – CSA C22.2 No. 107.1:16 (reaffirmed 2021)
- UL 1741 Supplement SB - Grid Support Utility Interactive Inverters and Converters Based upon IEEE 1547-2018 and IEEE 1547.1-2020 (Third Edition, Dated September 28, 2021)
- Photovoltaic (PV) DC Arc-Fault Circuit Protection-UL 1699B: First Edition, Dated August 22, 2018
- UL 1741 Supplement SA - Grid Support Utility Interactive Inverters and Converters (2010 Ed.2 Rev Feb. 15, 2018)
- CSA C22.2 NO. 330-23 (2023)

Reference standard

- IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces – IEEE Std 1547:2018 (Revision of IEEE Std 1547:2003), approved 15 February 2018; IEEE Std 1547a™-2020 Approved 9 March 2020. (Refer to IEEE Std 1547.1:2020, approved 5 March 2020.)
- Hawaiian Electric Companies, IEEE 1547.1-2020 Source Requirements Document Version 2.0 ("SRD V2.0"), effective on July 1st, 2020
- Generating Facility Interconnections- CPUC Rule 21 Effective on Jul 30, 2017
- UL1741 SECTION 92 TO SECTION 99, SEPTEMBER 28, 2021 - PHOTOVOLTAIC RAPID SHUTDOWN SYSTEMS
- UL 1741 (April 20, 2010 Ed.2) CRD, Subject: Power Control Systems (PCS)

Issuing Laboratory	SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch
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Page 2 of 74
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Product Type	Hybrid Solar Inverter
Trade Mark	
Product description	The products covered by this report are a permanently connected energy storage converter for use in electric power systems. It can invert the power from energy storage batteries and PV to grid or off grid port; It can also invert the power from grid to energy storage device. Unit is intended to be installed in indoor/outdoor location and not to be installed in hazardous location. The installation should accordance with the National Electrical Code, NFPA 70.
Model Number(s)	R8KLNA-P4, R12KLNA-P4, R16KLNA-P4, R8KLNA-P6, R12KLNA-P6, R16KLNA-P6
Model Differences	P4 series models have identical mechanical and electrical construction except some parameter configuration of the software architecture and quantity of non-critical components in order to control the max output power and some alternate components (refer to critical component list for details). P6 series models have identical mechanical and electrical construction except some parameter configuration of the software architecture and quantity of non-critical components in order to control the max output power and some alternate components (refer to critical component list for details). (*)The specific difference between P4 series models and P6 series models is that the wiring compartment of P4 series models does not have a circuit breaker, while the wiring compartment of P6 series models has input and output breakers.

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Electrical Ratings	R8KLNA-P4 PV side: 120-430V(500Vmax), 16A*4, 22A; Batteries side: 40-58V, 170A/170Amax AC grid side: 120Vac/240Vac Split phase, 60Hz, 33.3A(36.7Amax), -0.8~+0.8 EPS^(*) side: 120Vac/240Vac, Split phase, 60Hz, 8.8KVA(No PV), 8.8KVA(With PV), 33.3A(36.7Amax) R12KLNA-P4 PV side: 120-430V(500Vmax), 16A*4, 22A; Batteries side: 40-58V, 250A/260Amax AC grid side: 120Vac/240Vac Split phase, 60Hz, 50.0A(55.0Amax), -0.8~+0.8 EPS^(*) side: 120Vac/240Vac, Split phase, 60Hz, 12KVA(No PV), 13.2KVA(With PV), 50.0A(55.0Amax) R16KLNA-P4 PV side: 120-430V(500Vmax), 20A*4, 25A; Batteries side: 40-58V, 260A/280Amax AC grid side: 120Vac/240Vac Split phase, 60Hz, 66.7A(73.3Amax), -0.8~+0.8 EPS^(*) side: 120Vac/240Vac, Split phase, 60Hz, 13.2KVA(No PV), 13.2KVA(With PV), 50.0A(55.0Amax) R8KLNA-P6 PV side: 120-430V(500Vmax), 16A*4, 22A; Batteries side: 40-58V, 170A/170Amax AC grid side: 120Vac/240Vac Split phase, 60Hz, 33.3A(36.7Amax), -0.8~+0.8 EPS^(*) side: 120Vac/240Vac, Split phase, 60Hz, 8.8KVA(No PV), 8.8KVA(With PV), 33.3A(36.7Amax) R12KLNA-P6 PV side: 120-430V(500Vmax), 16A*4, 22A; Batteries side: 40-58V, 250A/260Amax AC grid side: 120Vac/240Vac Split phase, 60Hz, 50.0A(55.0Amax), -0.8~+0.8 EPS^(*) side: 120Vac/240Vac, Split phase, 60Hz, 12KVA(No PV), 13.2KVA(With PV), 50.0A(55.0Amax) R16KLNA-P6 PV side: 120-430V(500Vmax), 20A*4, 25A;
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	Batteries side: 40-58V, 260A/280Amax AC grid side: 120Vac/240Vac Split phase, 60Hz, 66.7A(73.3Amax), -0.8~+0.8 EPS(*) side: 120Vac/240Vac, Split phase, 60Hz, 13.2KVA(No PV), 13.2KVA(With PV), 50.0A(55.0Amax) (*) EPS: Equipment or facilities that deliver electric power to a load
Other Ratings.....:	Firmware Version: Ver 1.0406(DSP), Ver 2.0306(ARM) Normal operating performance Category B Abnormal operating performance Category III Type 3R, Grid Support Utility Interactive Converter Operating Temperature Range: -25~+60°C
Accessories.....:	NA
Testing Laboratory	Dongguan BALUN Testing Technology Co., Ltd.
Address.....:	Room 104, 204, 205, Building 1, No. 6, Industrial South Road, Songshan Lake District, Dongguan, Guangdong, China
Date of receipt of test item.....:	2024/01/26, 2024/02/22, 2024/07/25
Date(s) of performance of tests	2024/01/27 to 2024/03/27, 2024/02/22 to 2024/04/13, 2024/07/25 to 2024/07/29
Certification Conditions	N/A
(X Condition)	
Component Conditions	N/A
(U Condition)	

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Page 6 of 74
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CO-LISTING PAGE

CO-LISTING APPLICANT I

Applicant's Name:

Address:

Primary Contact:

Phone:

Email:

Product Correlation:

Basic Models	Co-Listed Models	Note any differences between Basic and Co-Listed models
1.		
2.		
3.		

CO-LISTING APPLICANT II

Applicant's Name:

Address:

Primary Contact:

Phone:

Email:

Product Correlation:

Basic Models	Co-Listed Models	Note any differences between Basic and Co-Listed models
1.		
2.		
3.		

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GENERAL INFORMATION**Components**

Components used in the covered products must be those outlined in the latest edition of the Listing Report.

Listing Marking

This report along with a valid certificate authorizes the certificate holder to use the Listing Marking of SGS North America only on products covered by this report and in accordance with the Product Certification Program Policy Handbook.

Production Line Tests

Manufacturing and Production Tests shall be performed as required by this Report.

Responsibility of the Manufacturer and Factory

- 1) It is the manufacturer and factory's responsibility to restrict the use of markings which reference SGS to those products which are found by the manufacturers own inspection to comply with the product description in this report. This includes reference to SGS directly and/or indirectly.
- 2) During hours in which the factory is in operation, the SGS inspector shall be given unlimited access to any portion of the premises where the product and/or parts are being produced, assembled, inspected and labeled; and to the test area designated for routine tests. The SGS inspector shall be permitted to inspect and subject the products to prescribed tests prior to shipment any product bearing or intended to bear marking referencing SGS.
- 3) The factory shall provide all required testing equipment and facilities including trained personnel for conducting all routine tests that are to be performed at the factory. These shall be available when needed so that the inspection work can proceed without delays.

Follow up Inspections

As part of the SGS Follow-Up inspection, it is required that an inspector periodically visit the factory location(s) and select for examination and/or testing, the most recent production sample of the product covered in the Listing Report.

SGS Inspector

- 1) A product which is found by the SGS inspector to have features which make it unacceptable to bear marking referencing SGS shall be corrected if the listing marking is to be used. The inspector shall carefully check additional production for such features until conditions are considered normal.
- 2) A product which does not comply with the provisions of the listing report shall have all reference to SGS removed. If the rejection of the product is questioned by the factory representative, it may be put on hold in separate area of location pending appeal. The factory shall satisfy the inspector that all marks referencing SGS are removed from the rejected material or finished product. Factory shall destroy product and or turn over to inspector for destruction.
- 3) All discrepancies between the product and the listing report shall be immediately stated to the attention of the factory representative. This shall be noted in the follow-up inspection report as a discrepancy.

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GENERAL CONSTRUCTIONAL REQUIREMENTS TO BE VERIFIED

Construction Details - This section specifies construction and component details in addition to the critical components table which are to be verified during factory follow-up inspections.

1. Mechanical Assembly – A wiring device shall be prevented from any turning that can apply tension to conductor connections, result in damage to the conductor, or otherwise adversely affect the assembly. Friction alone between the mating parts of the assembly shall not be acceptable to prevent turning.
2. Corrosion – All bare metal parts are protected against corrosion by coating, painting, plating or other means specifically identified in the specific construction details unless these metal parts inherently possess such properties to resist corrosion.
3. Accessibility of Live Parts – All uninsulated live parts in primary circuitry are housed within a metallic enclosure constructed without openings unless otherwise described in the construction details.
4. Grounding – All dead-metal parts external or within the enclosure that are exposed to contact during normal or any servicing operation are connected to the equipment grounding terminal.
5. Conductor Protection - Conductors that pass over edges or through openings in metal shall be secured from contacting the edges or be protected from cutting and abrasion. For sheet metal less than 1.1 mm thick, protection shall be provided by one of the following methods:
 - (a) rolling the edge of the metal not less than 120 degrees;
 - (b) a bushing or grommet of a material other than rubber at least 1.2 mm thick; or
 - (c) glass sleeving at least 0.25 mm thick.
6. Internal Wiring – UL/cUL listed, all wires in primary circuitry shall be type UL1015, minimum AWG 10, rated 600 V, 105 °C for AC side and UL1015 minimum AWG 12, rated 600V, 105°C for PV side unless otherwise described in this report.
7. Wire Connectors – UL/cUL listed, insulated closed-end type, PV:550V, 110°C, AC: 550V, 110°C, suitable for the wire size and number.
8. Insulating Materials - UL/cUL listed, rated 1500 V, V-0, suitable for the wire size and number.
9. Heat Shrinkable Tubing – UL recognized, rated 600 V, 125 °C, VW-1. Covered on wire connectors.
10. Marking and Labeling Systems – Pressure-sensitive labels and nameplates of the permanent type (Type P) that are secured by adhesive shall be in accordance with CSA C22.2 No. 0.15 or UL 969, suitable for damp locations and rated min. 60 °C when applied on Canopy.
11. Installation, Operating and Safety Instructions - Instructions for installation and use of this product are provided by the manufacturer as required by the standards.

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ROUTINE TESTS FOR MANUFACTURING AND PRODUCTION

The manufacturer shall perform the following Manufacturing and Production Tests on 100% of equipment produced which has both hazardous live parts and accessible conductive parts.

Required Tests

Dielectric Voltage Withstand Test
Utility Voltage and Frequency Variation Test
Grounding Continuity Test

Dielectric Voltage Withstand Test

Method

Each unit shall withstand without breakdown, as a routine production-line test, the application of a potential from Table 67.1A for AC rated circuits and from Table 67.1B for DC rated circuits:

- From input and output wiring, including connected components, to accessible dead metal parts that are able to become energized, and
- From input and output wiring to accessible low-voltage, limited-energy metal parts, including terminals.

Other than as noted in 67.3, the potential for the production-line test shall be in accordance with Condition A or Condition B of Table 67.1A or Table 67.1B at a frequency within the range of 40 – 70 Hertz.

Table 67.1A
Production-line test conditions AC rated circuits

Circuit rating, Vac	Condition A		Condition B		Condition C		Condition D	
	Potential, volts ac	Time, seconds	Potential, volts ac	Time, seconds	Potential, volts dc	Time, seconds	Potential, volts dc	Time, seconds
250 or less	1000	60	1200	1	1400	60	1700	1
More than 250	$1000+2 \sqrt{V^a}$	60	$1200+2.4 \sqrt{V^a}$	1	$1400+2.8 \sqrt{V^a}$	60	$1700+3.4 \sqrt{V^a}$	1

^a Maximum marked voltage.

Note: The multipliers in the table are chosen with the following background:

- 2.4 – A 20% adder on the multiplier 2 to account for reduced test time.
- 2.8 – $A \sqrt{2}$, truncated after the first decimal (≈ 1.4) multiplier on "2" from condition A to account for the peak value of an AC rms voltage to calculate the DC test potential of AC circuits.
- 3.4 – A combination of the two above: $2 \times 1.2 \times 1.4$, rounded to the next decimal.

Table 67.1B
Production-line test conditions for DC rated circuits

Circuit rating, Vdc	Condition A		Condition B		Condition C		Condition D	
	Potential, volts ac	Time, seconds	Potential, volts ac	Time, seconds	Potential, volts dc	Time, seconds	Potential, volts dc	Time, seconds
250 or less	1000	60	1200	1	1400	60	1700	1
More than 250	1000+1.4 V ^a	60	1200+ 1.7 V ^a	1	1400+ 2 V ^a	60	1700+2.4 V ^a	1

^a Maximum marked voltage.

A unit employing circuitry that is able to be damaged by an ac potential is able to be tested using a dc potential in accordance with Condition C or Condition D of Table 67.1A or Table 67.1B.

Testing of a unit in a heated or unheated condition meets the intent of the requirement for manufacturing and production tests.

The test is to be performed on a complete, fully assembled unit. It is not intended that the unit be unwired, modified, or disassembled for the test.

A unit employing a solid-state component that is not relied upon to reduce a risk of electric shock and that is susceptible to damage by the dielectric potential, is able to be tested before the component is electrically connected or after the component is electrically disconnected. The circuitry is able to be rearranged for the purpose of the test to minimize the potential of solid-state-component damage while retaining representative dielectric stress of the circuit.

Where the output rating of the test equipment transformer is less than 500 VA, the equipment is to include a voltmeter in the output circuit to directly indicate the test potential

Where the output rating of the test equipment transformer is 500 VA or more, the test potential is to be indicated:

- By a voltmeter in the primary circuit or in a tertiary-winding circuit,
- By a selector switch marked to indicate the test potential, or
- In the case of equipment having a single test-potential output, by a marking in a readily visible location to indicate the test potential. When marking is used without an indicating voltmeter, the equipment shall include a positive means, such as an indicator lamp, to indicate that the manually reset switch has been reset following a dielectric breakdown.

Test equipment, other than that described in 67.7 – 67.9, is usable when found to accomplish the intended factory control.

During the test, the unit switches are to be in the on position, both sides of the input and output circuits of the unit are to be connected together and to one terminal of the test equipment, and the second test-equipment terminal is to be connected to the accessible dead metal.

The results are acceptable if there is no dielectric breakdown.

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<u>Test Locations:</u>	<u>Test Voltage</u>	<u>Test Time</u>
L/N to metal enclosure	1700 Vdc for Dielectric Voltage withstand test	1 s
PV+/- to metal enclosure	2900 Vdc for Dielectric Voltage withstand test	1 s
DC+/- to metal enclosure	1700 Vdc for Dielectric Voltage withstand test	1 s
L/N to communication port	1700 Vdc for Dielectric Voltage withstand test	1 s
PV+/- to communication port	2900 Vdc for Dielectric Voltage withstand test	1 s
DC+/- to communication port	1700 Vdc for Dielectric Voltage withstand test	1 s

Utility Voltage and Frequency Variation Test

Method

As a routine production line test, each inverter shall be subjected to the Utility Voltage and Frequency Variation Test as per SB6 of UL Standard 1741 - 3rd Edition refer to IEEE 1547.1-2020, Section 7 (Production Tests).

Production tests shall include the following as applicable:

- Response to abnormal voltage (see 7.2)
- Response to abnormal frequency (see 7.3)

As a routine production line test, each utility-interactive inverter initially exporting power within its normal operating range shall cease to export power to the simulated utility source after the output voltage and frequency of the simulated utility source are adjusted to each condition specified below within the time as noted in the Table 13 and Table 18 (extracted from IEEE 1547-2018). The inverter is to be tested to at each default condition once to verify compliance.

Table 13—DER response (shall trip) to abnormal voltages for DER of abnormal operating performance Category III (see Figure H.9)

Shall trip function	Shall trip—Category III			
	Default settings ^a		Ranges of allowable settings ^b	
	Voltage (p.u. of nominal voltage)	Clearing time (s)	Voltage (p.u. of nominal voltage)	Clearing time (s)
OV2	1.20	0.16	fixed at 1.20	fixed at 0.16
OV1	1.10	13.0	1.10–1.20	1.0–13.0
UV1	0.88	21.0	0.0–0.88	21.0–50.0
UV2	0.50	2.0	0.0–0.50	2.0–21.0

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Table 18 —DER response (shall trip) to abnormal frequencies for DER of abnormal operating performance Category I, Category II, and Category III (see Figure H.10)

Shall trip function	Default settings ^a		Ranges of allowable settings ^b	
	Frequency ^c (Hz)	Clearing time (s)	Frequency (Hz)	Clearing time (s)
OF2	62.0	0.16	61.8–66.0	0.16–1 000.0
OF1	61.2	300.0	61.0–66.0	180.0–1 000.0
UF1	58.5	300.0 ^c	50.0–59.0	180.0–1 000
UF2	56.5	0.16	50.0–57.0	0.16–1 000

Each inverter with field adjustable trip points shall have the trip factory set points confirmed in accordance with the manufacturer's installation instructions.

The inverter is not required to be tested at full output power and Signal injection methods shall be permitted to be used for production testing.

WARNING: The factory test(s) specified may present a hazard of injury to personnel and/or property and should only be performed by persons knowledgeable of such hazards and under conditions designed to minimise the possibility of injury.

Grounding Continuity Test

Method

The impedance at 60 hertz between the point of connection of the equipment-grounding means and any other metal part that is required to be grounded shall not be more than 0.1 ohm, the voltage when a current of 40A amperes derived from a 60-hertz source with a no-load voltage not exceeding 6 volts is passed between the grounding connection and the metal part in question.

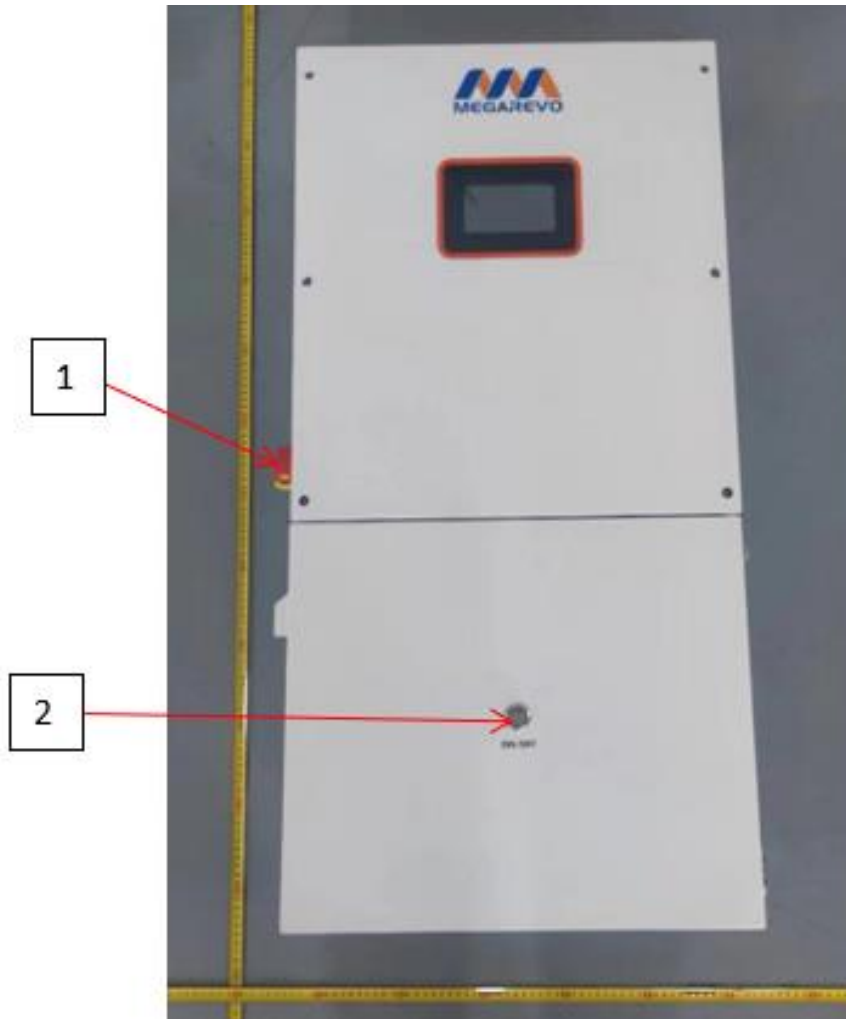
A visual or audible device (ohmmeter, buzzer, etc.) may be used to indicate grounding continuity.

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 13 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

PHOTOGRAPHS

Photo 1: Overall view of enclosure (Hybrid Solar Inverter, model: R12KLNA-P6)

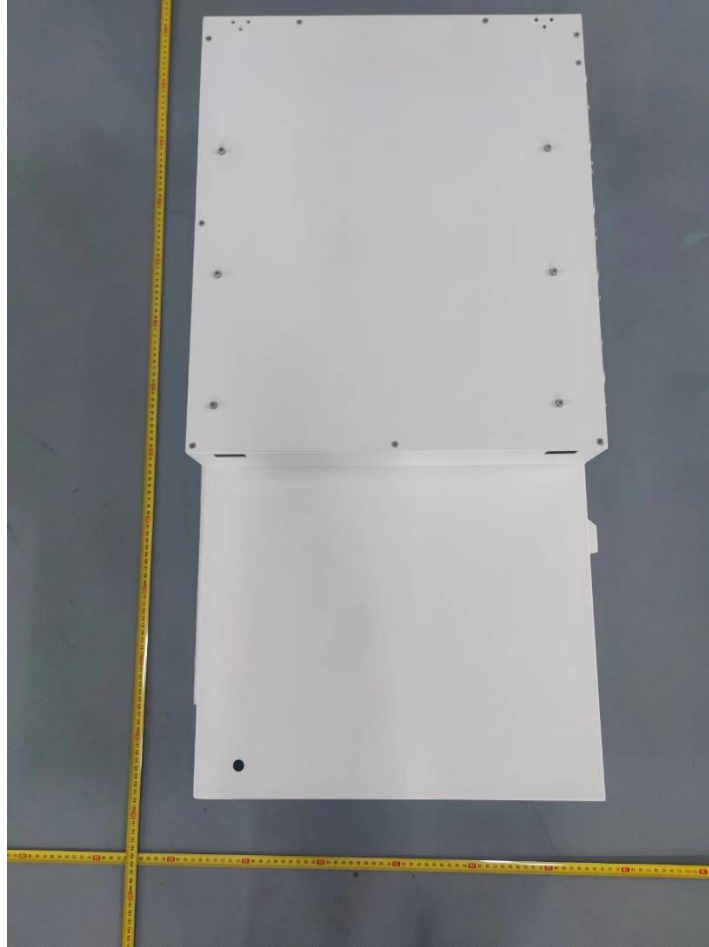


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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 14 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 2: Rear view of enclosure (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 15 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 3: Upper cover face (Hybrid Solar Inverter, model: R12KLNA-P6)



Photo 4: Back of upper cover face (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 16 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 5: Back of face (Hybrid Solar Inverter, model: R12KLNA-P6)

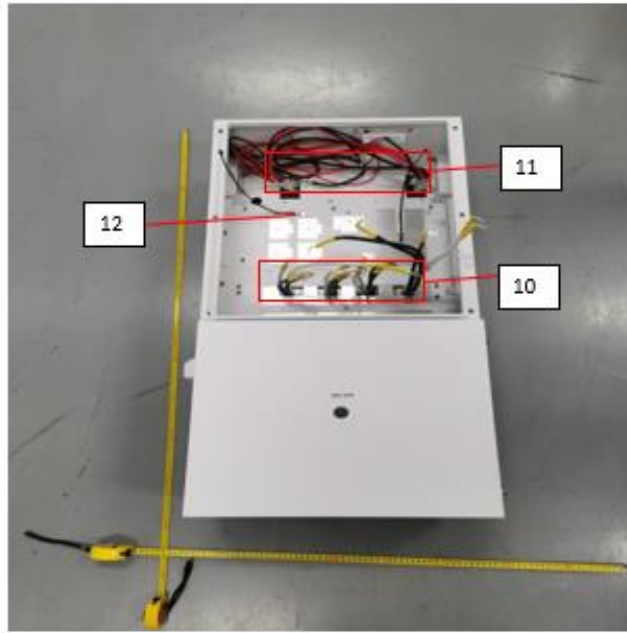
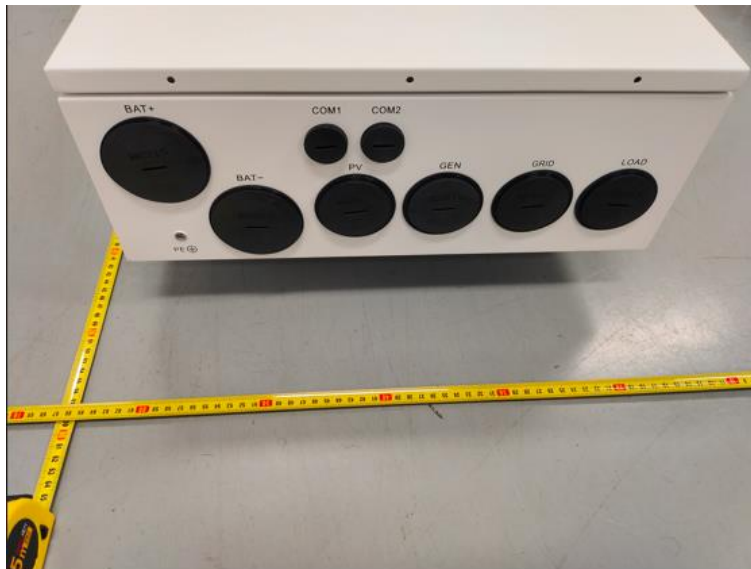


Photo 6: Connection interface (Hybrid Solar Inverter, model: R12KLNA-P6)



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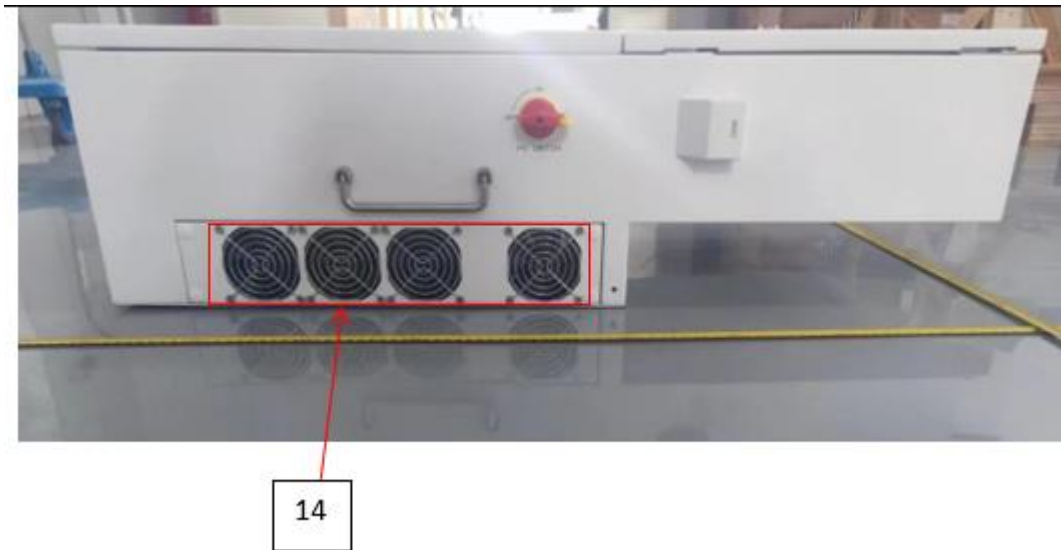
SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 17 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 7: Right side view of enclosure (Hybrid Solar Inverter, model: R12KLNA-P6)



Photo 8: Left side view of enclosure (Hybrid Solar Inverter, model: R12KLNA-P6)



14

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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 18 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 9: Top view of enclosure (Hybrid Solar Inverter, model: R12KLNA-P6)

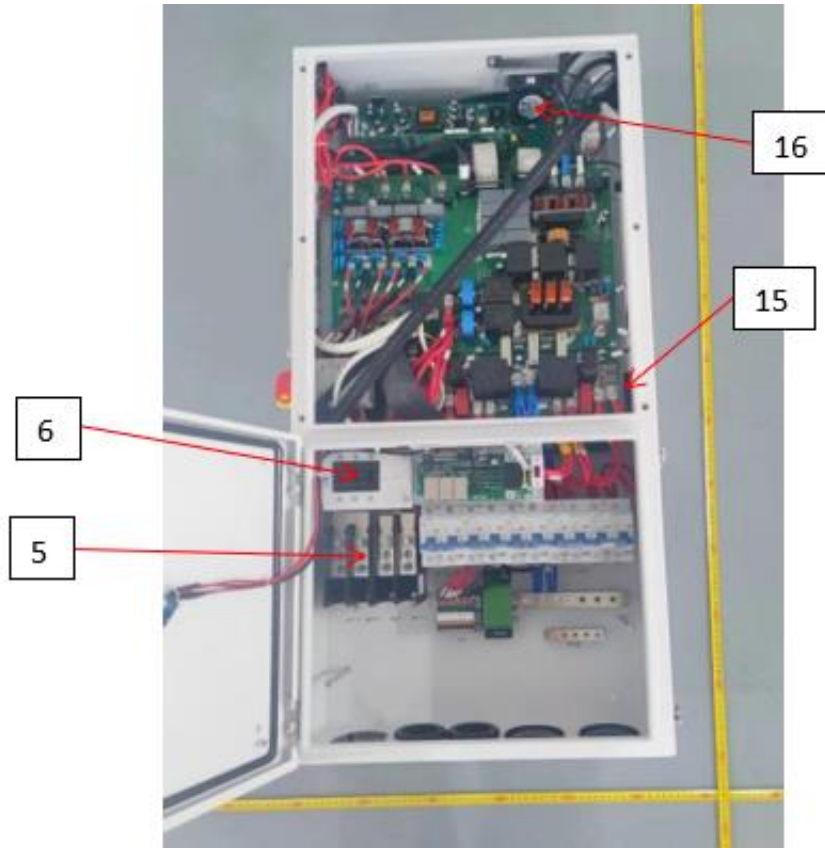


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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 19 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 10: Internal view of unit (Hybrid Solar Inverter, model: R12KLNA-P6)

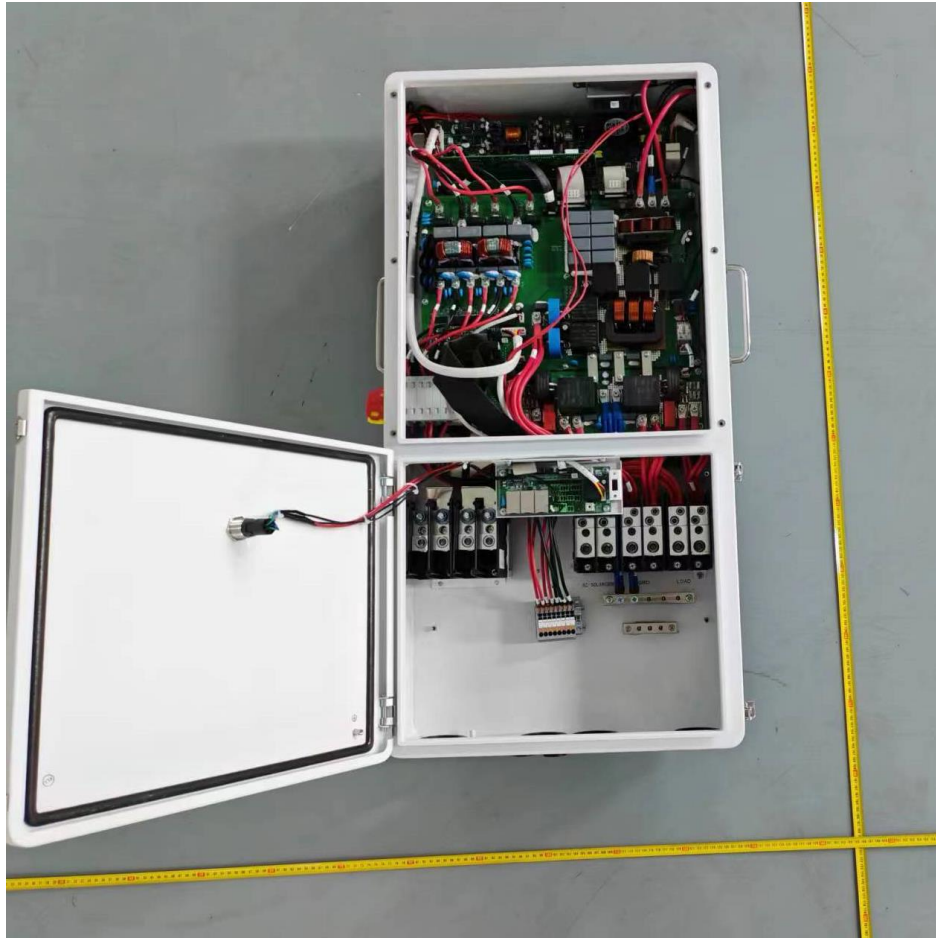


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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 20 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 11: Internal view of unit (Hybrid Solar Inverter, model: R12KLNA-P4)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 21 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 12: DR2 board (Hybrid Solar Inverter, model: R12KLNA-P6)

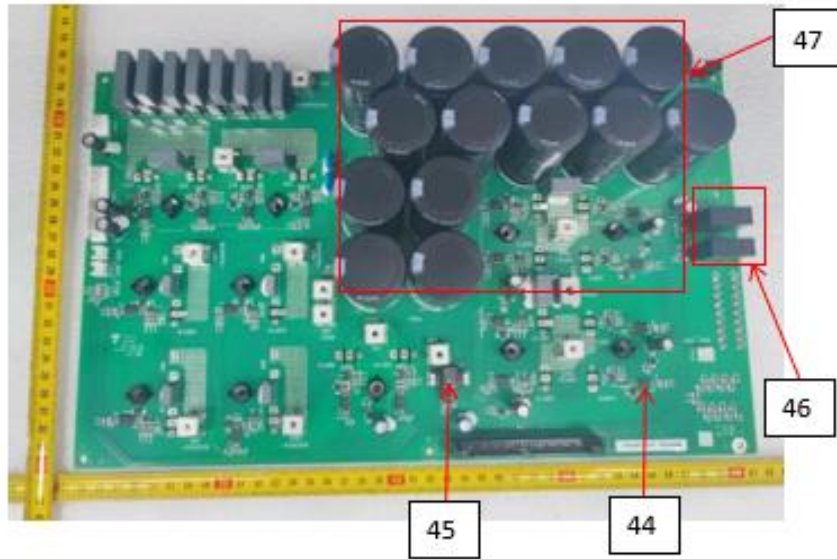
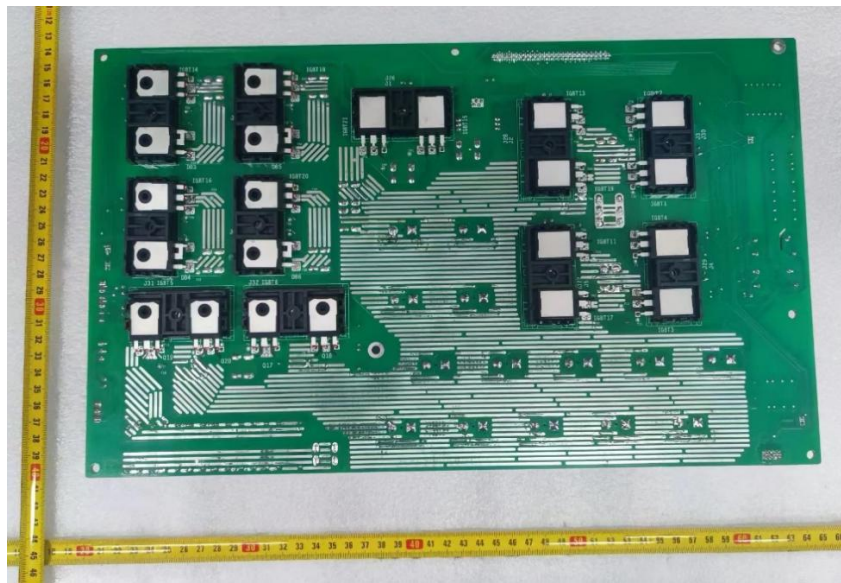


Photo 13: Back view of DR2 board (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 22 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 14: DR2 board (Hybrid Solar Inverter, model: R12KLNA-P6)

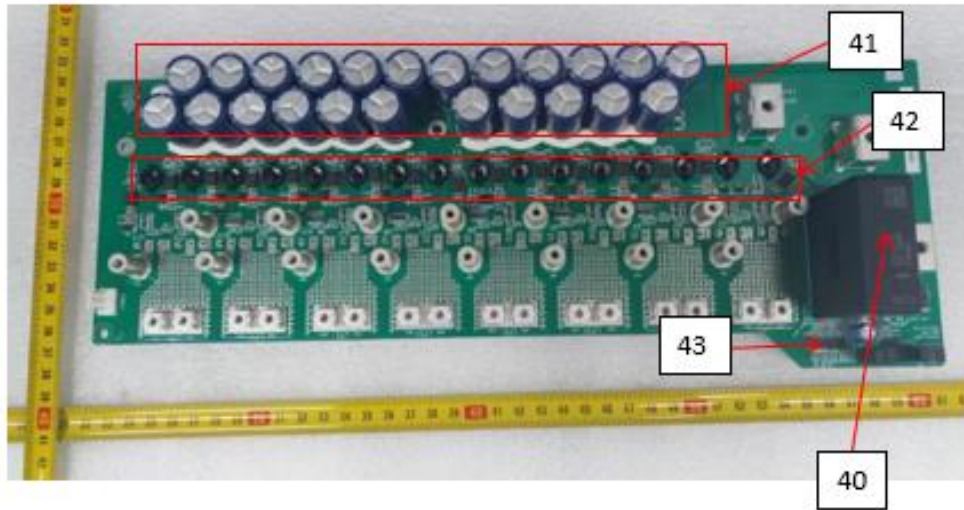
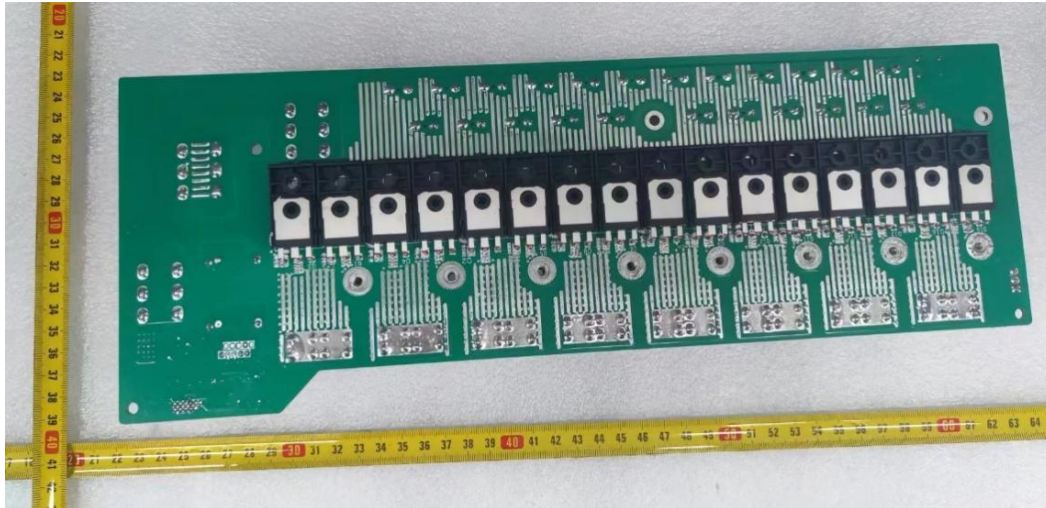


Photo 15: Back view of DR2 board (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 23 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 16: EM4 board (Hybrid Solar Inverter, model: R12KLNA-P6)

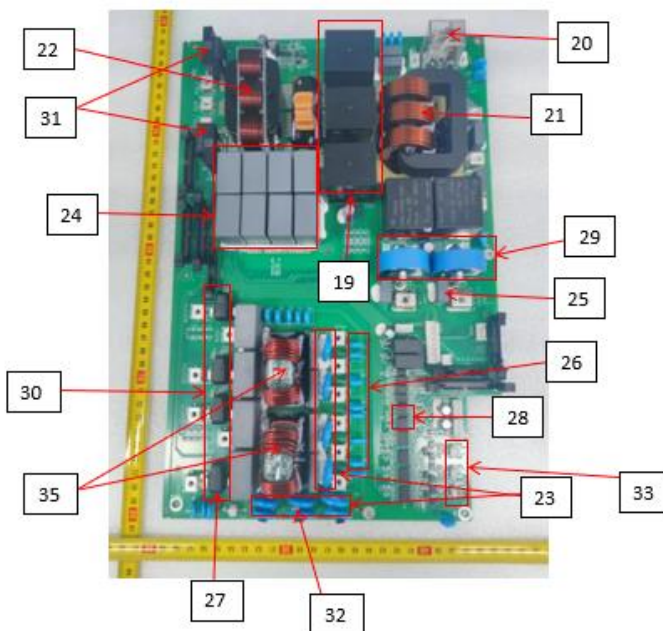
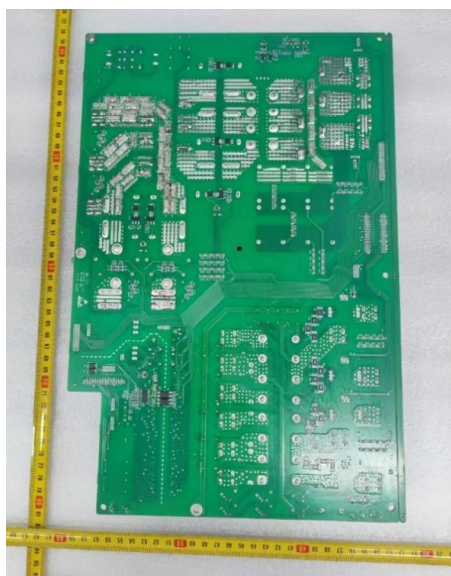


Photo 17: Back view of EM4 board (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 24 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 18: Control panel board (Hybrid Solar Inverter, model: R12KLNA-P6)

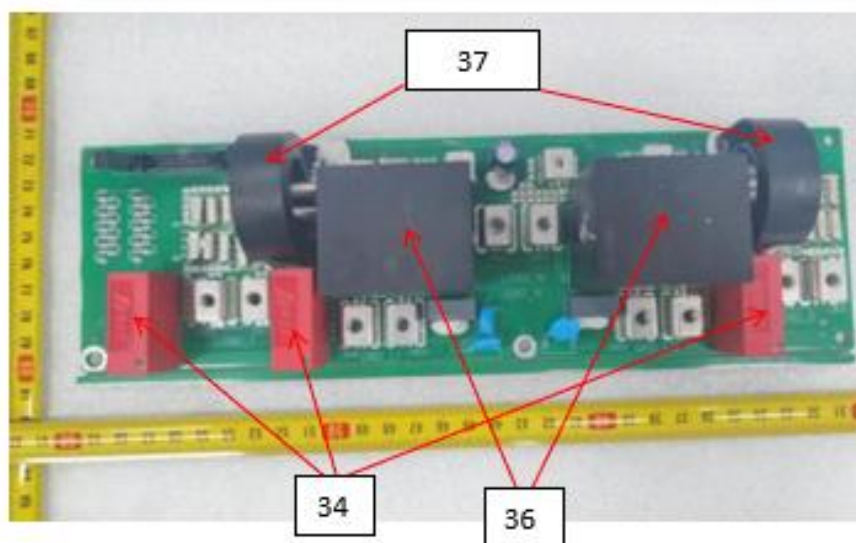
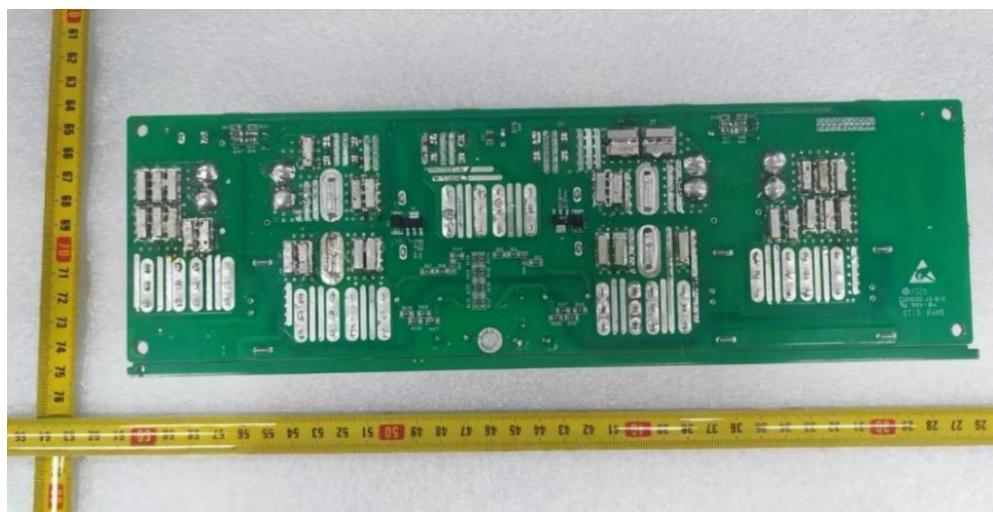


Photo 19: Back view of Control panel board (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 25 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 20: PW3 board (Hybrid Solar Inverter, model: R12KLNA-P6)

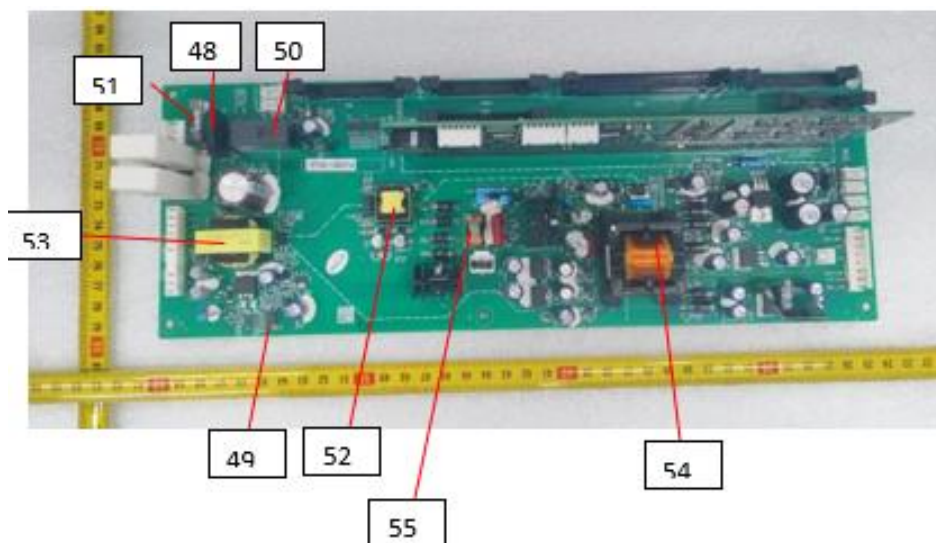
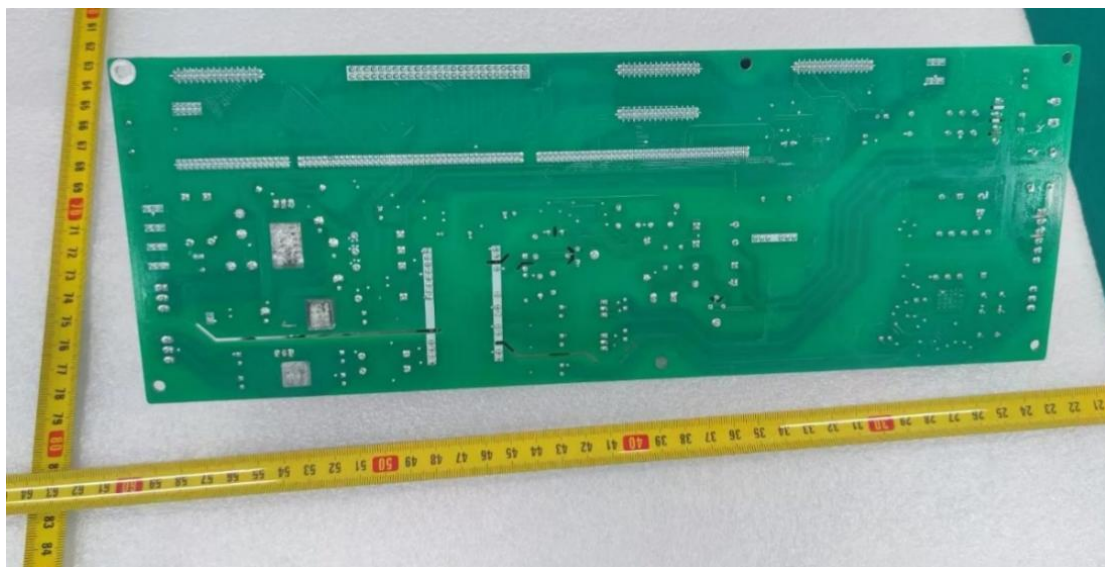


Photo 21: Back view of PW3 board (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 26 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 22: Communication Board (Hybrid Solar Inverter, model: R12KLNA-P6)

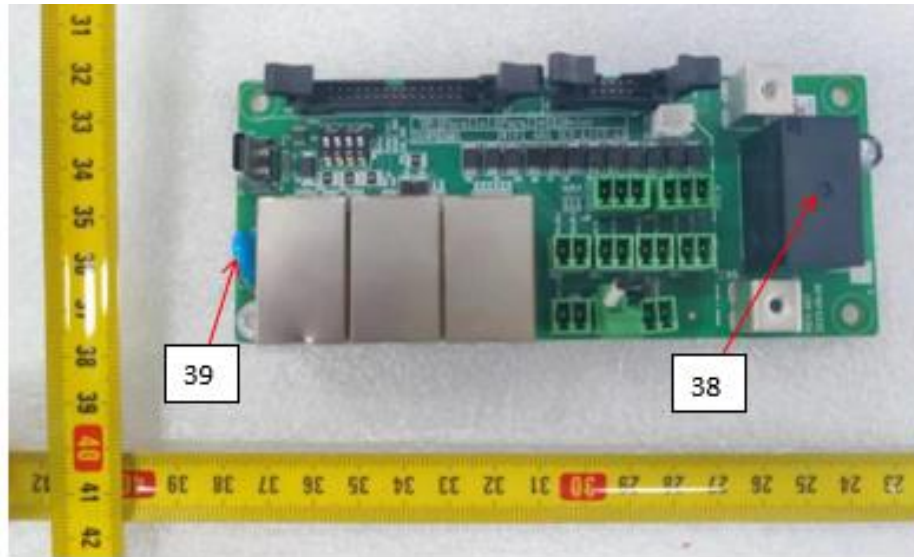
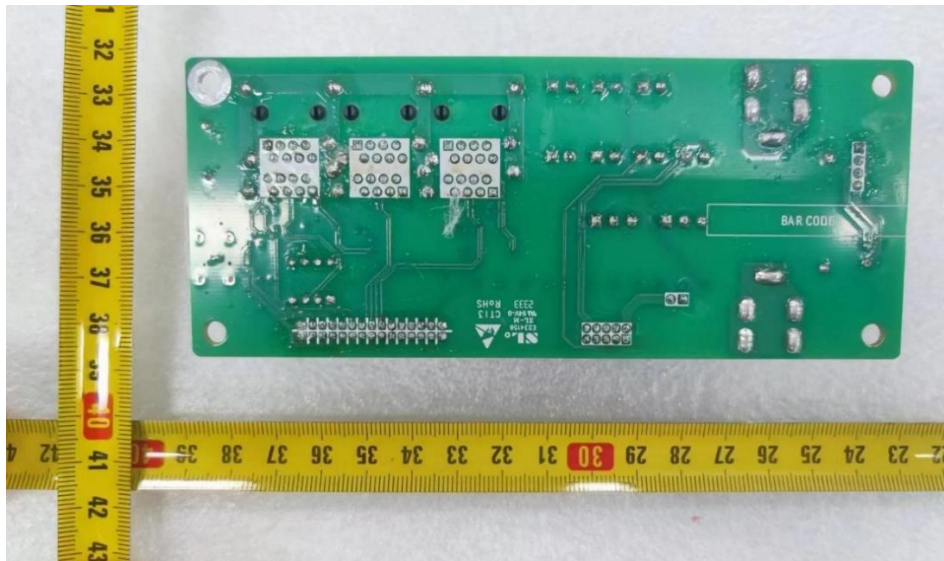


Photo 23: Back view of Communication Board (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 27 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 24: display screen (Hybrid Solar Inverter, model: R12KLNA-P6)



Photo 25: Back view of display screen (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 28 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 26: Power grid load diesel generator connection port (Hybrid Solar Inverter, model: R12KLNA-P6)

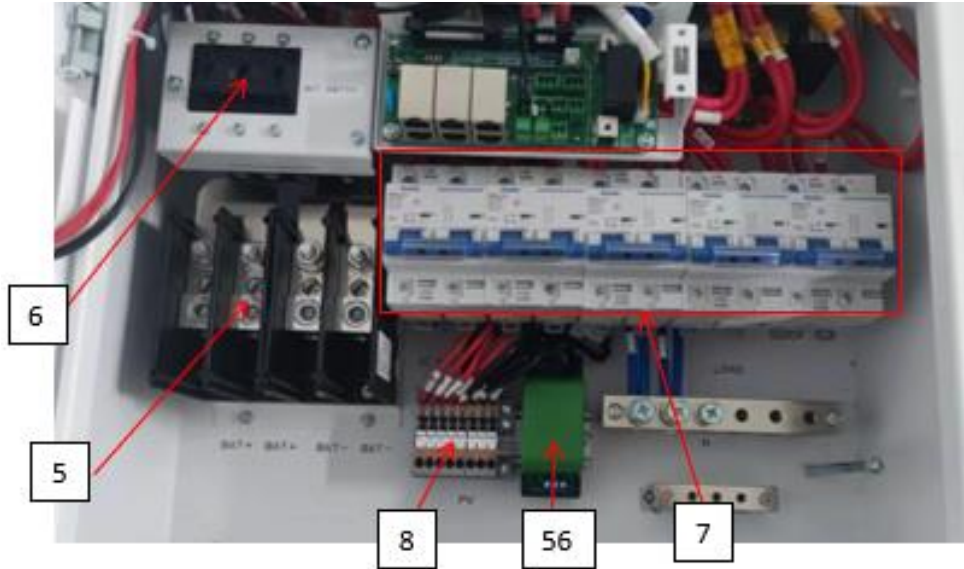
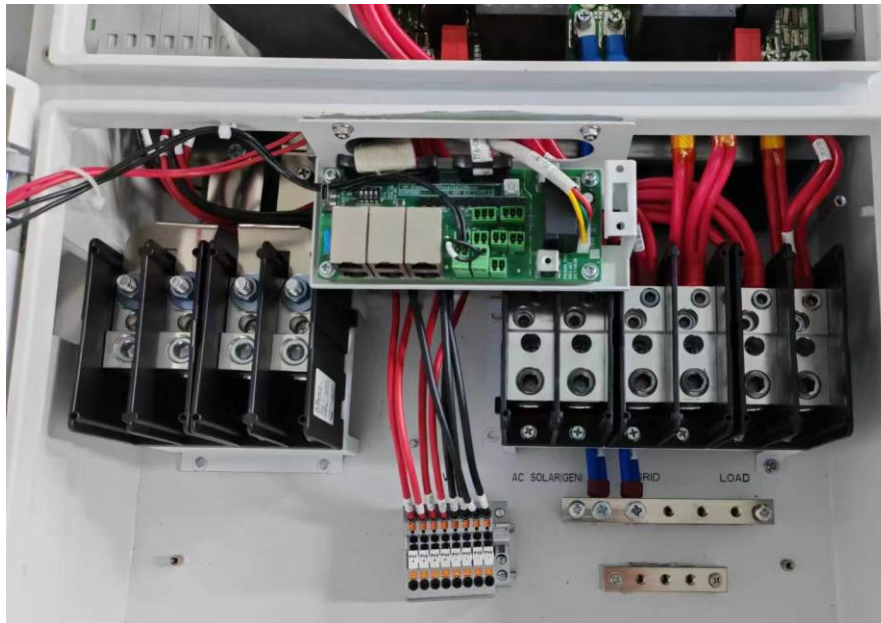


Photo 27: Power grid load diesel generator connection port (Hybrid Solar Inverter, model: R12KLNA-P4)



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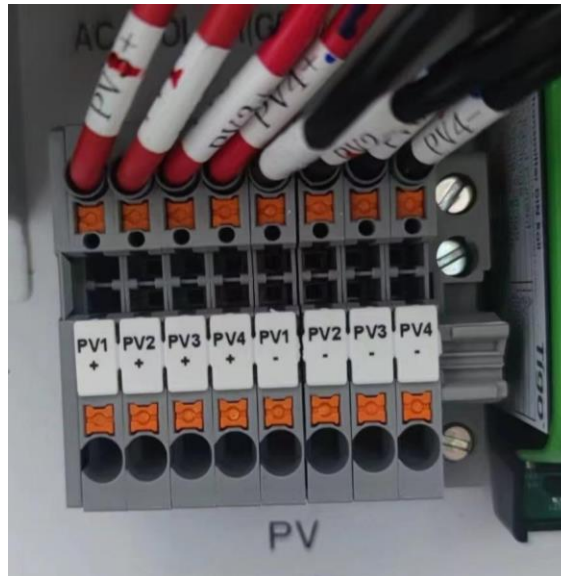
SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 29 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 28: Battery connection port (Hybrid Solar Inverter, model: R12KLNA-P6)



Photo 29: PV wiring port (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 30 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 30: port (Hybrid Solar Inverter, model: R12KLNA-P6)



Photo 31: grounding (Hybrid Solar Inverter, model: R12KLNA-P6)



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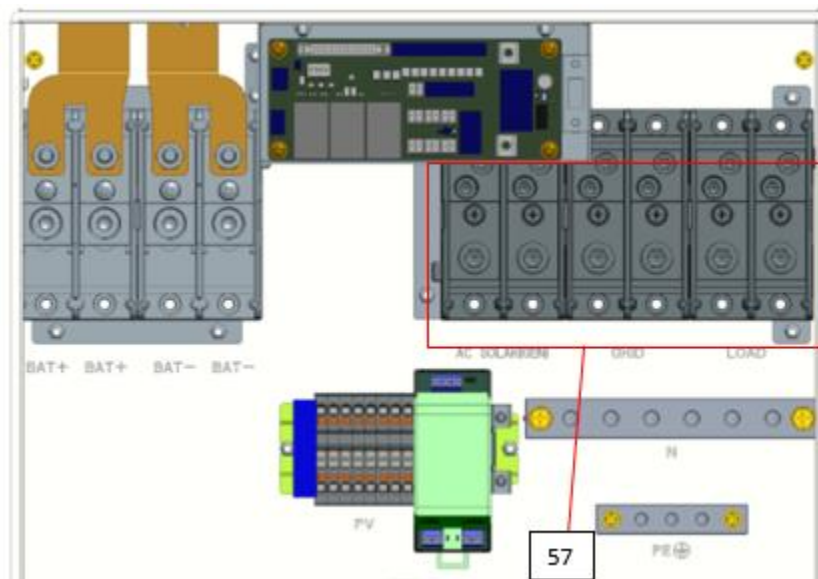
SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 31 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 32: Arc detection module (Hybrid Solar Inverter, model: R12KLNA-P6)



Photo 33: AC out terminal (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 32 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 34: PV switch (Hybrid Solar Inverter, model: R12KLNA-P6)

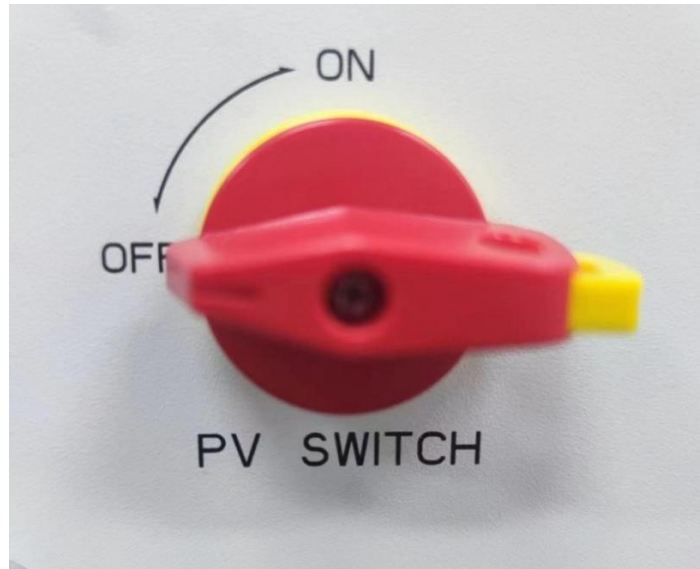


Photo 35: internal fan (Hybrid Solar Inverter, model: R12KLNA-P6)



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 33 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Photo 36: Wire compartment (Hybrid Solar Inverter, model: R12KLNA-P6)

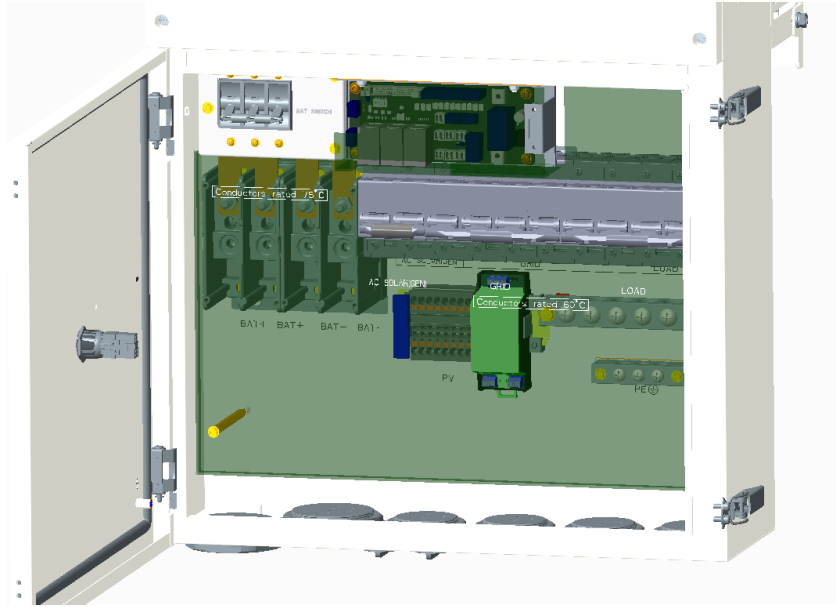
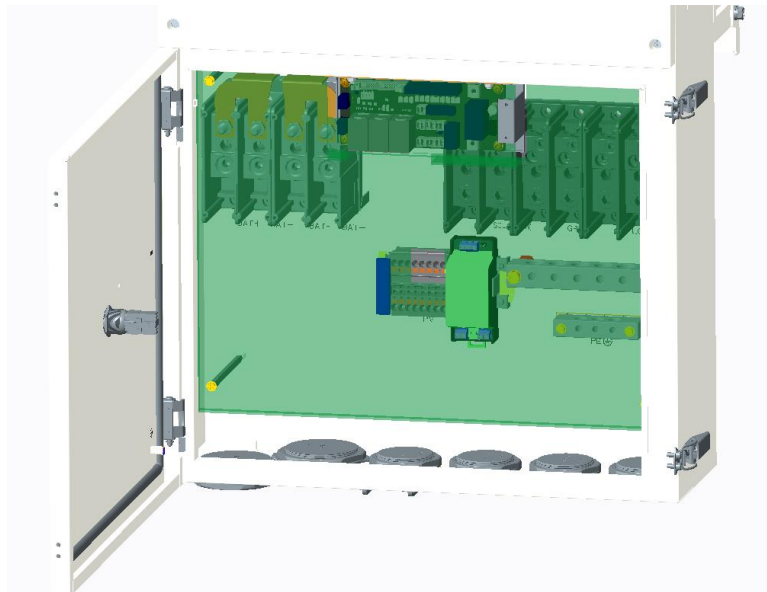


Photo 37: Wire compartment (Hybrid Solar Inverter, model: R12KLNA-P4)



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CRITICAL COMPONENTS

The following components are considered “critical” in terms of this certification and must be verified during factory inspections. No substitutions or alternate components are allowed unless specifically as stated in this report.

Photo No.	Item No.	Component name	Manufacturer/ trademark	Type/model	Technical data	Standard	Mark(s) of conformity
1	1	DC switch	Santon International BV	XBUE3610-6-D	600Vdc/25Adc/6P , -40°C~85°C	ANSI/UL 508	UR
		Alternative	Su zhou Puzhao new energy Equipment Co., LTD	PEDS150R-HM32-6	600Vdc/25Adc/6P , -40°C~85°C	ANSI/UL 508	UR
		Alternative	Su zhou Puzhao new energy Equipment Co., LTD	PEDS150H-HM50-6	800Vdc/63Adc/6P , -40°C~85°C	ANSI/UL 508	UR
	2	button switch	Wen zhou Jin hong Electric Appliance Co., Ltd	J19-372YWT	250Vac /5A , -25°C~55°C	CSA C22.2	cURus
--	3	Sealant	RAMPF Polymer Solutions GmbH & Co KG	HT-820	V-0, HF-1, 130°C	ANSI/UL 94	UR
--	4	Potting glue in inductive metal box	SHEN ZHEN ANPIN SILICONE MATERIAL CO LTD	905	V-0, 150°C	ANSI/UL 94, UL746A, UL746B	cURus
10	5	Battery terminal block (optional)	Tai Wan He Bi HOPPY	ATH-255-1, ATH-255-3, ATH-255-4	600V/255A, -20°C~60°C	UL1059, UL486E	cURus
	6	Battery switch	Zhe Jiang BSB Electrical Appliances Co Ltd	B3T1-250.0/80-3600	80V 250A -40°C~85°C	UL1077, UL489A, UL489	UR

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 35 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

		Alternative	Zhe Jiang BSB Electrical Appliances Co Ltd	B3T1- 300.0/80- 3600B-D2-C1- G-H	80V 300A -40°C~85°C	UL1077, UL489A, UL489	RT
26	7	AC switch	Zhe Jiang BSB Electrical Appliances Co Ltd	B3T1- 100.0/250-260	250V 125A -40°C~85°C	UL1077, UL489A, UL489	UR
		Alternative	Nader	NDB1-125 C125-2P	250V 125A, - 40°C~70°C	UL1077	UR
		Alternative	Zhe Jiang BSB Electrical Appliances Co Ltd	B3T1- 100.0/125-260	250V 100A, -40°C~85°C	UL1077, UL489A, UL489	UR
		Alternative	Nader	NDB3-100	250V 100A, -40°C~70°C	UL1077	UR
	8	PV terminal	Phoenix contact Gmbh& Co	UP-PV8-PT4	800V 32A,- 60°C~110°C	CSA- C22.2 No. 158	UR
		Alternative	Phoenix contact Gmbh& Co	UP-PV8-PT6	1000V 41A,- 60°C~110°C	CSA- C22.2 No. 158	UR
	9	AC terminal	Shen Zhen Connection Electronics Co.,Ltd	DRTB150-BR- 02	2PIN;600V;300A, -40°C~120°C	CSA- C22.2 No. 158	UR
		Alternative	Shen Zhen Connection Electronics Co.,Ltd	DRTB50-BR	2PIN;600V 150A,- 40°C~120°C	CSA- C22.2 No. 158	UR
		Alternative	Tai Wan He Bi HOPPY	ATH-255	600V/255A, - 20°C~60°C	UL1059, UL486E	cURus
	10	LLC Transformer materials					
5	10-1	COPPER WIRE	PACIFIC ELECTRIC WIRE &CABLE(SHE NZHEN)CO LTD	MW79-C (UEFN/U)	Polyurethane Copper Wire , 155°C	UL1446	UR
		Alternative	TAI- IELECTRIC WIRE &	MW79-C (UEFN/U)	Polyurethane Copper Wire , 155°C	UL1446	UR

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 36 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

			CABLE CO LTD				
		Alternative	TAYA ELECTRIC WIRE & CABLE CO LTD	MW79-C (UEFN/U)	Polyurethane Copper Wire , 155°C	UL1446	UR
		Alternative	ROSHOW TECHNOLOGY CO LTD	MW79-C (UEFN/U)	Polyurethane Copper Wire , 155°C	UL1446	UR
		Alternative	FENG CHING METAL CORP	MW79-C (UEFN/U)	Polyurethane Copper Wire , 155°C	UL1446	UR
		Alternative	WEIFENG	(MW79-C (UEFN/U)	155°C	UL1446	UR
		Alternative	DAYI	MW79-C (UEFN/U)	155°C	UL1446	UR
	10-2	TAPE	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD	TYPE NO.: PF-301	Polimide film tape 180°C	UL1446	UR
	10-3	SILICONE TUBE	SHENZHEN WAHCHANG WEI INDUSTRIAL CO LTD	SRS-70	Rating Thermal 200°C	UL1446	UR
		Alternative	KING TAI LAI	SRC	Rating Thermal 200°C	UL1446	UR
	10-4	NOMEX	SUI ON INSULATING MATERIALS CO LTD	TJ6640 NMN	180°C	UL1446	UR
		Alternative	JINLONG	6641 DMD#\$@-F	155°C	UL1446	UR

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 37 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

10-5	VARNISH	KYOCERA ChEMICAL CORP	TVB2180T* a	130 /155°C	UL1446	UR
10-6	Heat Shrink Tube	CHANGYUAN ELECTRONIC S(SHENZHEN)CO LTD	CHANGBAO1 02	125°C	UL1446	UR
	Alternative	WOER	RSFR	125°C	UL1446	UR
10-7	MARGIN TAPE	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD	Nonwoven Cloth TYPE NO.:WF	Rating Thermal 130°C	UL1446	UR
10-8	EPOXY	DONGGUAN EATTO	E500(XX)	130°C	UL1446	UR
10-9	Silica gel	HuiTian	5297	1.5 W/m·K,- 40°C~180°C	UL1446	UR
11	Power Inductance materials					
11-1	Tape	JINGJIANG PRESSURE SENSITIVE GLUE FTY	PF 0.06	180°C	UL1446	UR
11-2	Film	E I DUPONT DE NEMOURS & CO INC	200C(f1)	200°C	UL1446	UR
11-3	MYLAR	TORAY INDUSTRIES INC	Lumirror X-10	130°C	UL1446	UR
11-4	Varnish	ZHUHAI CHANGX IANNEW MATERLALS TECHNOLOG Y CO LTD	E962	130°C	UL1446	UR
	Alternative	WU JIANG TAIHU INSULATING MATERIAL CO LTD	ET-90(a)	130°C	UL1446	UR
	Alternative	ZHUHAI CHANGX	E57(a)	130°C	UL1446	UR

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			IANNEW MATERLALS TECHNOLOG Y CO LTD				
	11-5	EPOXY	ZHONGSHAN SHUNXIN CHEMICAL CO LTD	SX1166A/B	105°C	UL1446	UR
		Alternative	HEN ZHEN ANPIN SILICONE MATERIAL CO LTD	906A/B	105°C	UL1446	UR
	12	NTC Temp sensor	SHEN ZHENSHI KEMIN SENSOR CO LTD	CWF4 15KJ4150 L=250MM	φ4.0*24/L=250MM/ , -30°C~120°C	CSA LTR No. I-003	cURus
32	13	Arc detection module	Fonrich (Shang Hai) New Energy Technology Co., Ltd	FR-DCMG- AS4A	81*42*43mm/±15A , -40°C~70°C	UL1699 B	cCSAus
8	14	External fan	NMB	08025VE- 12M-CLD	12V/0.3A/ - 10°C~65°C 80*80mm	CSA- C22.2 No. 113	cUR
		Alternative	PROTECHNI C ELECTRIC	MGT6012VB- R25_D	12V/0.34A/45dB/, -10°C~70°C /80*80mm	CSA- C22.2 No. 113	cUR
10	15	Circulating fan at low voltage side of battery	NMB	06025SA12N AL01	12V/0.2A/44dB/, - 10°C~65°C /60*60mm	CSA- C22.2 No. 113	cUR
		Alternative	PROTECHNI C ELECTRIC	MGT6012XB- R25	12V/0.12A/33.9dB/ , -10°C~70°C /60*60mm	CSA- C22.2 No. 113	cUR
	16	Circulating fan	PROTECHNI C ELECTRIC	MGT6012XB- R25	12V/0.12A/33.9dB/ , -10°C~70°C /60*60mm	CSA- C22.2 No. 113	cUR
		Alternative	NMB	06025SA12N AL01	12V/0.2A/44dB/, -10~65°C /60*60mm	CSA- C22.2 No. 113	cUR

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--	17	PCB material	GANZHOU BANGDE CIRCUIT TECHNOLOGY CO LTD	BD-D	V-0, 130°C	ANSI/UL 94; ANSI/UL 796 ;CAN/CSA-C22.2 No. 0.17	cURus
--	18	Power grid CT	Yuan Xing International Inc	CTSA035-200A/100mA B2	2000:1;0.5;-40°C~75°C	UL2808	UR
16	19	Relay(RLY1)	Xia Men Hongfa Electroacoustic Co.,Ltd.	HF167F-12-HTF	90A;40 0Vac, -40°C~85°C	ANSI/UL 508; CAN/CSA C22.2 No. 14-13	UR
		Alternative	Dong guan Churod Electronic Co., Ltd	CHAR-112A90C	90A;400Vac, -40°~85°C	ANSI/UL 508; CAN/CSA C22.2 No. 14-13	UR
	20	Relay (K1)	Xia Men Hongfa Electroacoustic Co.,Ltd.	HF18FF/A230-3Z2T	230AC;10A;AC;230;10A;21.5*28*36#21.0*28*35;-40°C~70°C	UL 508, CAN/CSA C22.2 No. 14/05	cURus
		Alternative	Zhe Jiang Chint Electrics Co Ltd	JZX-22F AC220V-3Z1	230AC;10A, -25°C~55°C	UL 508, CAN/CSA C22.2 No. 14/05	UR
	21	Common mode choke materials (L15)					
	21-1	WIRE	JUNG SHING WIRE CO LTD	PEW	155°C	UL94	UR
		Alternative	TAI-I ELECTRIC WIRE & CABLE CO LTD	UEWF @	155°C	UL94	UR
		Alternative	PACIFIC ELECTRIC WIRE & CABLE	PEWF/U	155°C	UL94	UR

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 40 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

			(SHENZHEN) CO LTD				
		Alternative	XINGNING JINYAN ELECTRICAL CO LTD	xPEW	155°C	UL94	UR
		Alternative	HOI LUEN ELECTRICAL MFR CO LTD	xPEW	155°C	UL94	UR
		Alternative	DONGGUAN YIDA ENDUSTRIAL CO., LTD	*PEW	155°C	UL94	UR
	21-2	BOARD	KINGBOARD LAMINATES HOLDINGS LTD	22.0mm,FR-4,	130°C	UL94	UR
	21-3	EPOXY	DONG GUAN SHI PAI HUA CHUANG MATERIAL FTY	808A/B	130°C	UL94	UR
		Alternative	DONGGUAN EATTO ELECTRONIC MATERIAL CO LTD	3300A/B	130°C	UL94	UR
	22	Differential Mode Choke materials (L16)					
	22-1	WIRE	JUNG SHING WIRE CO LTD	2*6mm	200°C	UL94	UR
		Alternative	TAI-I ELECTRIC WIRE & CABLE CO LTD	2*6mm	200°C	UL94	UR
		Alternative	PACIFIC ELECTRIC WIRE & CABLE (SHENZHEN) CO LTD	2*6mm	200°C	UL94	UR
		Alternative	XINGNING JINYAN	2*6mm	200°C	UL94	UR

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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 41 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

			ELECTRICAL CO LTD				
		Alternative	HOI LUEN ELECTRICAL MFR CO LTD	2*6mm	200°C	UL94	UR
		Alternative	DONGGUAN YIDA ENDUSTRIAL CO., LTD	2*6mm	200°C	UL94	UR
	22-2	VARNISH	WU JIANG TAIHU INSULATING MATERIAL CO LTD	T-4260(a)	130°C	UL94	UR
	22-3	TAPE	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD	CT-280	130°C	UL94	UR
	22-4	SPACER	KINGBOARD LAMINATES	94V-0 2.0&2.5mm	155°C	UL94	UR
		Alternative	FOSHAN NAN HAI CHANGSHI ELECTRONIC MATERIAL CO.,LTD	FR-4	2.0&2.5mm , 155°C	UL94	UR
	22-5	EPOXY	DONGGUAN SHIPAI HUACHUANG MATERIAL FACTORY	808A/B	130°C	UL94	UR
	22-6	INSULATION	ISOVOLTA	Voltaflex DMD3 0180	180°C	UL94	UR
	23	Varistor (RV1- RV10)	Bright king	621KD20-UL	620V;23*23*6.1*10 ;UL/CSA/VDE;RoH S, -40°C~85°C	UL1449	UR
	24	X2 Capacitance (C172)	Xia Men faratronic Co Ltd	C42Q2475KB AC000	4.7uF±10%- 305Vac-32*33*18- PIN=27.5mm- 305V;RoHS, - 40°C~110°C	UL1414	UR

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	25	X2 Capacitance (C31)	Shen Zhen surong capacitors Co.,Ltd	MP3334KHD4 G0	305Vac;X2;0.33uF; 10%, - 40°C~110°C	UL1414	UR
		Alternative	Xia Men faratronic Co Ltd	C42Q2334K6 SC000	305Vac;X2;0.33uF; 10%, - 40°C~110°C	UL 60384-14	UR
	26	Chip safety capacitor (C1-21,23,)	Izumo murata manufacturing co., Ltd	DE1E3KX472 MA4BL01	4700pF±20%- 400VAC, - 25°C~125°C	UL 60384-14	UR
	27	Ceramic gas discharge tube (GDT1)	EPCOS	A71-H14X	600V 5KA;UL;RoHS, - 40°C~90°C	UL1449	UR
		Alternative	SHEN ZHEN BEN CENT ELECTRONIC CO LTD	B8G600M	600V 5KA;UL;RoHS, - 40°C~85°C	UL1577	UR
	28	CAN Transceiver(U7-8)	Texas Instruments	ISO1050DUB R	CAN transceiver SOP- 8;TI#ISO1050DUB R-UL, -55°C~ 105°C	UL1577	UR
	29	Current Transformer (L3-L4)	Yuan Xing International Inc	TA22E11- 100A/50mA	2000:1;-40°C~ 75°C;E; 100A/50mA;RoHS	UL2808	UR
	30	Current Sensor (SN1-SN4)	LEM electronics China Co.,Ltd	HLSR 20-P	20 A;5V;-40°C~ 105°C;RoHS	UL508:20 10	UR
	31	Current Sensor (SN5)	Sinomags technology Co.,Ltd	STB-100LA/S	100 A; ±0.625V;5V;0.5µs;- 40°C~ 105°C;RoHS	UL508:20 10	UR
	32	Fuse (F2-F7)	Shen Zhen Yabao electronics Co.,Ltd	A8-3A-F	250V;3A;6.2*2.5*6. 3, 150°C	UL 60691	UR
	33	Gas discharge tube (GDT3-5)	Tai wan Brightking	3RL090M-5-S	90V;5kA;5*7.5mm; SMD, -40°C~ 85°C	ANSI/UL 497B	UR

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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 43 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

18	34	Surge protection device (F8-10)	Anhui Jinli electric technology Co.,Ltd	GTSP-MDV670/20/A	670VDC;20kA;10kA;35*14*27mm, -40℃~85℃	UL1449 UL94 V-0	UR
16	35	PV inductance (L1 L2)					
	35-1	TAPE	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD	CT-280	130℃	UL94	UR
	35-2	BOARD	KINGBOARD LAMINATES HOLDINGS LTD	FR-4	1.5mm , 130℃	UL94	UR
	35-3	TUBE	CHANGYUAN ELECTRONIC S(SHENZHEN) CO LTD	CB-TT-S	200℃	UL94	UR
	35-4	VARNISH	WU JIANG TAIHU INSULATING MATERIAL CO LTD	ET-90(a)	180℃	UL94	UR
		Alternative	ELANTAS ELECTRICAL INSULATION ELANTAS PDG INC	468-2(x)	180℃	UL94	UR
EM5 board							
18	36	Relay (K2-3)	Dong Guan Churod Electronic Co., Ltd	CHAR-112A200	200A;830VAC;12V DC;-40℃~85℃	ANSI/UL5 08;CAN/ CSA/C22. 2No.14-13	UR
		Alternative	Xia Men Hongfa Electroacousti c Co.,Ltd.	HF172F-200/12-HTF	200A;830VAC;12V DC;-40℃~85℃	ANSI/UL5 08;CAN/ CSA/C22. 2No.14-13	UR
	37	Current transformer(TR1-2)	YuanXing International Inc	CTT0250B	200A/100mA,-40℃~85℃	UL57.13	UR
TF6 board							

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 44 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

--		Relay (K1)	Xia Men Hongfa Electroacoustic Co.,Ltd.	HF161F-40W	43A 277VAC, -40°C~85°C	ANSI/UL508;CAN/CSA/C22.2No.14-13	UR
22	38	Alternative	ZETTLER	AZSR143-1AE-12D(103)	50A 277VAC, -40°C~85°C	ANSI/UL508;CAN/CSA/C22.2No.14-13	UR
		Alternative	Dong Guan Churod Electronic Co., Ltd	CHFN-V-112H1A2F	50A 277VAC, -40°C~85°C	ANSI/UL508;CAN/CSA/C22.2No.14-13	UR
	39	Safety capacitor (C2)	Murata	DE1E3KX472 MA4BL01	4700pF±20%-400VAC(UL250VAC), -25°C~125°C	UL 60384-14	UR
DR2 board							
14	40	Current Sensor (SN2)	Sinomags technology Co.,Ltd	STB-300LA/ZNH	300A;85°1.1;100mA;5V;0.6us;-40°~105°	UL508:2010	UR
	41	Electrolytic (C391)	Hu Nan aihua group Co.,Ltd	ERJ1JM122M 30OTR	63V;1200uF;±20%;18*31.5mm;-40°C~105°C;7000h@105°C;2.8A	--	--
		Alternative	Nichicon	UPW1J122M HD	63V;1200uF;±20%;18*31.5mm;-40°C~105°C;7000h@105°C;2.8A	--	--
	42	Optocoupler (PC19-34)	Nichicon corporation	NSi6801B-DSWFR	28ns;25ns;5000V;SOIC-6, -40°C~125°C	UL1577	cURus
		Alternative	TI	UCC23513BD WYR	28ns;25ns;5000V;SOIC-6, -40°C~125°C	UL1577	cURus
	43	Optocoupler (PC39)	Toshiba Corporation	TLP7820	1.7µs;4000V;SSO-8;RoHS,-40°C~105°C	UL1577	UR
		Alternative	Texas Instruments	AMC1301DW VR	1.7µs;4000V;SSO-8;-40°C~125°C	UL1577	UR
	DR3 board						

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 45 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

12	44	Optocoupler (PC1)	Texas Instruments	UCC23513D WYR	28ns;25ns;5000V;S OIC-6,-40℃~130℃	UL1577	UR
		Alternative	Novosense Microelectronics	NSI6801C-DSWFR	28ns;25ns;5000V;S OIC-6,-40℃~150℃	UL1577	UR
	45	Current Sensor (SN1)	LEM electronics China Co.,Ltd	HLSR 32-P	32 A;2.5V;+5v;2.5us;8 5℃;RoHS	UL508:20 10	UR
	46	Relay (K1-2)	Xia Men Hongfa Electroacoustic Co.,Ltd.	HF36F/012-HST-UL	12VDC/10A250VAC,-30℃~70℃	ANSI/UL508;CAN/CSA/C22.2No.14-13	cURus
		Alternative	TYCO	SDT-SH-112DM-UL	12VDC/10A250VAC,-30℃~70℃	ANSI/UL508;CAN/CSA/C22.2No.14-13	UR
	47	Bus capacitor(C313)	Hu Nan aihua group Co.,Ltd	ELT2FM152 R60KT	315VDC,1500UF,-25℃~105℃	--	--
		Alternative	Nantong Jianghai Capacitor Co.,Ltd	ECS2FBB1 52MLA35006 0V	315VDC,1500UF,-25℃~105℃	--	--
PW3 board							
20	48	Rectifier bridge (DB1)	LESHAN RADIO COMPANY, LTD	GBU8K	1000V;8A;1V, -55℃ ~ +150℃	UL94	UR
	49	Optocoupler (PC1-3)	SHARP	PC123X5YI P0F-UL	CTR200% ~ 400%;-30℃ ~ 100℃	UL1577	UR
		Alternative	Tai wan sheng Lite on Group	LTV-816S-TP-C-CU-UL	CTR200% ~ 400%;-30℃ ~ 110℃	UL1577	cURus
	50	Relay (RLY1)	Xia Men Hongfa Electroacoustic Co.,Ltd.	HF115F/01 2-2ZS4AF	8A/240VAC, -25℃ ~ +85℃	ANSI/UL 508; CAN/CSA C22.2. No. 14-10	UR

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 46 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

		Alternative	SONG CHUAN PRECISION CO LTD	507N-2CH-F-C12VDC	8A/240VAC, -25°C ~ +85°C	ANSI/UL 508; CAN/CS A C22.2. No. 14-10	UR
51		Fuse (F1)	COOPER BUSSMANN LLC	S505H-1-R	250Vac,5*20mm/-40°C ~ +85°C	UL248-1/14,CSA 248-1/14	UR
		Alternative	LITTLFUSE	04770001.M XP	250Vac,5*20mm/-55°C ~ 125°C	UL248-1/14,CSA 248-1/14	UR
52	High-frequency transformer(TR1)						
52-1	BOBBIN		CHANG CHUN PLASTICS CO.,LTD	PHENOLIC T375HF EF-16 (7+7PIN)	150°C	UL1446	UR
52-2	COPPER WIRE		HUIZHOU GOLDEN OCEAN MAGNET WIRE FACTORY	XUEW-F MW79C	155°C	UL1446	UR
	Alternative		HOI LUEN ELECTRICAL MFR CO.,LTD	XUEW-F MW79C	155°C	UL1446	UR
52-3	TRIPLE WIRE		GREAT LEOFLON INDUSTRIAL CO LTD	TRW(B)*	130°C	UL1446	UR
52-4	TAPE		HUIZHOU YA HUA STICKING TAPE CO.,LTD	CT	130°C	UL1446	UR
52-5	BUSHING		SHEN ZHEN WOER HEAT SHRINKABLE MATERIAL CO.,LTD	TEFLON/W F	200°C	UL1446	UR
53	High-frequency transformer(TR3)						
53-1	BOBBIN		CHANG CHUN PLASTICS CO.,LTD	PHENOLIC T375HF,	150°C	UL1446	UR

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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 47 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

	53-2	COPPER WIRE	HUIZHOU GOLDEN OCEAN MAGNET WIRE FACTORY	XUEW-F MW79C,	155°C	UL1446	UR
		Alternative	HOI LUEN ELECTRICAL MFR CO.,LTD	XUEW-F MW79C,	155°C	UL1446	UR
	53-3	TAPE	HUIZHOU YA HUA STICKING TAPE CO.,LTD	CT,	130°C	UL1446	UR
	53-4	TUBE	SHEN ZHEN WOER HEAT SHRINKABLE MATERIAL CO.,LTD	TEFLON/W F,	200°C	UL1446	UR
	53-5	MARGIN TAPE	HUIZHOU YA HUA STICKING TAPE CO.,LTD	WF,	130°C	UL1446	UR
	53-6	EPOXY Dark glue	DONGGUAN SHIPAI HUACHUANG MATERIAL FACTORY	H907-HF	-60°C ~ 150°C	UL1446	UR
	53-7	VARNISH	SUZHOU TAIHU INSULATING MATERIAL CO LTD	T-4260(a),	130°C	UL1446	UR
	54	High-frequency transformer(TR4)					
	54-1	Wire	HENG YA ELECTRIC(D ONGGUAN) TD	Φ0.30mm MW75C	155°C	UL1446	UR
		Alternative	PACIFIC ELECTRIC WIRE&CABL E(SHENZHEN GCO.,LTD.	Φ0.30mm MW75C	155°C	UL1446	UR
		Alternative	TAI-I ELECTIC	Φ0.30mm MW75C	155°C	UL1446	UR

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 48 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

			WIRE&CABL E.CO.LTD				
54-2	Tape		JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO., LTD	PF*(d)(g)	180°C	UL1446	UR
54-3	Tube		CHANGYUAN ELECTRONIC S (SHENZHEN) CO LTD	CB-TT-L	200°C/150V	UL1446	UR
54-4	TRIPLE WIRE		E&B TECHNOLOG Y CO LTD	E&B -B Φ0.50mm	130°C/1.41kVpeak	UL1446	UR
54-5	MARGIN TAPE		JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO.,LTD	WFF(f)	130°C	UL1446	UR
54-6	Varnish		HITACHI CHEMICAL CO LTD	WP-2952F- 2G	155°C	UL1446	UR
54-7	Bobbin		SUMITOMO BAKELITE	PM-9820, PM-9630	150°C/94V-0	UL1446	UR
55	Common mode choke(L3)						
55-1	Varnish		ELANTAS ELECTRICAL INSULATION ELANTAS PDG INC	468-2(x)	155°C	UL1446	UR
	Alternative		WU JIANG TAIHU INSULATING MATERIAL CO LTD	ET-90(a)	155°C	UL1446	UR
55-2	BOARD 1.0		KINGBOARD LAMINATES	KB-3150N,	130°C	UL1446	UR
	Alternative		KINGBOARD LAMINATES	KB-6150,	130°C	UL1446	UR
55-3	BOARD 0.5		KINGBOARD LAMINATES	KB-3150N,	130°C	UL1446	UR

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		Alternative	KINGBOARD LAMINATES	KB-6150,	130°C	UL1446	UR
	55-4	Adhesive	DONGGUAN EATTO ELECTRONIC MATERIAL CO LTD	3300A-1/3300B-1	150°C	UL1446	UR
		Alternative	DONGGUAN EATTO ELECTRONIC MATERIAL CO LTD	E-500 (XX)	150°C	UL1446	UR
26	56	Complete Machine RSD Switch					
	56-1	Transmitter PCBA, Single Rss Core	TIGO	490-00100-10	12V/1A, - 40°~ 85 °C°	UL1741	UR
	56-2	Transmitter PCBA, Dual Rss Core (Alternative)	TIGO	490-00100-20	12V/1A, - 40°~ 85 °C°	UL1741	UR
	56-3	Transmitter Din Rail, Single Rss Core (Alternative)	TIGO	490-00100-10	12V/1A, - 40°~ 85°C	UL1741	UR
	56-4	Transmitter Din Dual Rss Core (Alternative)	TIGO	490-00100-20	- 40°~ 70°C	UL1741	UR
33	57	AC output terminal	Tai Wan He Bi HOPPY	ATH-255-2, ATH-255-5	600V/255A, - 20°C~60°C	UL1059, UL486E	UR
36, 37	58	Barrier	SABIC INNOVATIVE PLASTICS US LLC	FFBOARD	2mm, 120°C	ANSI/UL94	UR

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 50 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

ANNUAL RE-TESTING OF UNLISTED COMPONENTS

The unlisted components on this page are uncontrolled (not falling under a third party certification program) and require periodic retesting and/or evaluation.

Note to SGS Follow Up Inspector: The inspection office will notify you in writing when these components must be selected and sent to the Lab indicated below for re-evaluation.

Ship the samples to:

Ship the samples to SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch

198 Kezhu Road, Science City, Economic & Technology Development Area,
Guangzhou, Guangdong, China

The unlisted components covered by this report and are required to be re-tested/evaluated are shown in the following table:

PHOTO #		ITEM #	DESCRIPTION	MFR	TYPE/ MODEL	RATING
N/A		N/A	N/A	N/A	N/A	N/A
VERIFICATION PROCESS						
Test Standard:		N/A				
Frequency of Testing:		N/A	N/A	N/A		
Clause No.	TEST		PARAMETERS			
N/A	N/A		N/A			

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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 51 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

MARKINGS

The following markings are required:

Marking A: Nameplate

 MEGAREVO Hybrid Solar Inverter R8KLNA-P4 Grid Support Interactive Inverter	
PV Input	
Max PV Input	12kW
Mppt input Voltage	120~430V
MAX.Input Current	16A*4
MAX.Short Circuit Current	22A*4
AC Output (On Grid)	
Rated Output Power	8kVA
Rated Output current	33.3A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8
Max.Grid Passthrough Current	200A
AC Output (EPS)	
Rated Output Power	8kVA
Rated Output Current	33.3A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz
Battery	
Battery Voltage Range	40V~58V
Max. charging Current	170A
Max. Discharging Current	170A
System	
Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	55Kg
Max. Efficiency	96.5%
PV DC AFCI	Type 1
PVRSS	Tigo RSS transmitters (RS2)
Product Serial number	
   	
S/N	
Shenzhen Megarevo Technology Co., Ltd.	

 MEGAREVO Hybrid Solar Inverter R12KLNA-P4 Grid Support Interactive Inverter	
PV Input	
Max PV Input	18kW
Mppt input Voltage	120~430V
MAX.Input Current	16A*4
MAX.Short Circuit Current	22A*4
AC Output (On Grid)	
Rated Output Power	12kVA
Rated Output current	50A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8
Max.Grid Passthrough Current	200A
AC Output (EPS)	
Rated Output Power	12kVA
Rated Output Current	50A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz
Battery	
Battery Voltage Range	40V~58V
Max. charging Current	250A
Max. Discharging Current	260A
System	
Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	55Kg
Max. Efficiency	96.5%
PV DC AFCI	Type 1
PVRSS	Tigo RSS transmitters (RS2)
Product Serial number	
   	
S/N	
Shenzhen Megarevo Technology Co., Ltd.	



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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 52 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Marking A: Nameplate

 MEGAREVO Hybrid Solar Inverter R16KLNA-P4 Grid Support Interactive Inverter	
PV Input	
Max PV Input	24kW
Mppt input Voltage	120~430V
MAX.Input Current	20A*4
MAX.Short Circuit Current	25A*4
AC Output (On Grid)	
Rated Output Power	16kVA
Rated Output current	66.7A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8
Max.Grid Passthrough Current	200A
AC Output (EPS)	
Rated Output Power	13kVA
Rated Output Current	54.2A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz
Battery	
Battery Voltage Range	40V~58V
Max. charging Current	260A
Max. Discharging Current	280A
System	
Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	55Kg
Max. Efficiency	96.5%
PV DC AFCI	Type 1
PVRSS	Tigo RSS transmitters (RS2)
Product Serial number	
   	
S/N	
Shenzhen Megarevo Technology Co., Ltd.	

 MEGAREVO Hybrid Solar Inverter R8KLNA-P6 Grid Support Interactive Inverter	
PV Input	
Max PV Input	12kW
Mppt input Voltage	120~430V
MAX.Input Current	16A*4
MAX.Short Circuit Current	22A*4
AC Output (On Grid)	
Rated Output Power	8kVA
Rated Output current	33.3A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8
Max.Grid Passthrough Current	200A
AC Output (EPS)	
Rated Output Power	8kVA
Rated Output Current	33.3A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz
Battery	
Battery Voltage Range	40V~58V
Max. charging Current	170A
Max. Discharging Current	170A
System	
Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	56Kg
Max. Efficiency	96.5%
PV DC AFCI	Type 1
PVRSS	Tigo RSS transmitters (RS2)
Product Serial number	
   	
S/N	
Shenzhen Megarevo Technology Co., Ltd.	

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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 53 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Marking A: Nameplate



Hybrid Solar Inverter R12KLNA-P6 Grid Support Interactive Inverter

PV Input

Max PV Input	18kW
Mppt input Voltage	120~430V
MAX.Input Current	16A*4
MAX.Short Circuit Current	22A*4

AC Output (On Grid)

Rated Output Power	12kVA
Rated Output current	50A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8

Max.Grid Passthrough Current 200A

AC Output (EPS)

Rated Output Power	12kVA
Rated Output Current	50A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz

Battery

Battery Voltage Range	40V~58V
Max. charging Current	250A
Max. Discharging Current	260A

System

Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	56Kg
Max. Efficiency	96.5%
PV DC AFCI	Type 1
PVRSS	Tigo RSS transmitters (RS2)

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.



Hybrid Solar Inverter R16KLNA-P6 Grid Support Interactive Inverter

PV Input

Max PV Input	24kW
Mppt input Voltage	120~430V
MAX.Input Current	20A*4
MAX.Short Circuit Current	25A*4

AC Output (On Grid)

Rated Output Power	16kVA
Rated Output current	66.7A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8

Max.Grid Passthrough Current 200A

AC Output (EPS)

Rated Output Power	13kVA
Rated Output Current	54.2A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz

Battery

Battery Voltage Range	40V~58V
Max. charging Current	260A
Max. Discharging Current	280A

System

Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	56Kg
Max. Efficiency	96.5%
PV DC AFCI	Type 1
PVRSS	Tigo RSS transmitters (RS2)

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.

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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 54 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Marking B: Cautionary

	WARNING -Hot surfaces. -To reduce the risk of burns. -Do not touch. AVERTISSEMENT -Surfaces chaudes. -Réduire les risques de brûlures. -Ne touchez pas.
	WARNING -Both AC and DC voltage sources are terminated inside this equipment. -Each circuit must be individually dis-connected before servicing. -When the Photovoltaic array is exposed to light, it supplies a DC voltage to this equipment. AVERTISSEMENT -Les sources de tension alternative et continue sont raccordées à l'intérieur de cet équipement. -Chaque circuit doit être déconnecté avant de procéder à toute réparation. -Lorsque le réseau photovoltaïque est exposé à la lumière, il fournit une tension continue à cet équipement. WARNING -Disconnect all sources of supply before servicing. AVERTISSEMENT -Couper toutes les sources d'alimentation avant de faire l'entretien et les réparations. WARNING -When a ground fault is indicated. Normally grounded conductors may be ungrounded and energized or normally ungrounded conductors may be grounded. AVERTISSEMENT -Lorsqu'un défaut à la terre est signalé, Les conducteurs normalement mis à la terre peuvent ne pas être mis à la terre et être sous tension ou les conducteurs qui, Normalement, ne sont pas mis à la terre peuvent être mis à la terre.
	WARNING -Risk of electric shock from energy stored in capacitor. -Do not remove cover until 5 minutes after disconnecting all sources of supply. AVERTISSEMENT -Risque d'électrocution en raison de l'électricité emmagasinée. -Attendez 5 minutes après avoir débranché toutes les sources d'alimentation avant d'y accéder.
	CAUTION -Risk of electric shock, Do not remove upper front cover. -No user serviceable parts inside. -Refer servicing to qualified service personnel. ATTENTION -Risque d'électrocution, Ne pas enlever le capot avant supérieur. -Aucune utilisateur pièce réparable à l'intérieur. -Référez le service à personnel de service qualifié. Transformerless inverter, Non-isolated inverter. Inverseur sans transformateur, onduleur non-isolé. Interactive PCE. Appareillage de conversion de puissance interactif.

	Risk Of Electric Shock. Risque de choc électrique. Use Copper Conductors Only. Utiliser uniquement des conducteurs en cuivre
	Use conductor with insulation rated for at least 90 °C. Utilisez un conducteur avec une isolation nominale d'au moins 90 °C. Reference to the instruction manual for the tightening torque to be applied to the wiring terminals. Référence au manuel d'instructions pour le couple de serrage à appliquer aux bornes de câblage.

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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 55 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Note:

1. Marking A shall be permanent. SGS NA Listed Mark shall be at least 5 mm high. Other letters shall be at least 2.4 mm high.
2. Marking B shall be permanent, and A cautionary marking shall be prefixed by the word "CAUTION", "WARNING", or "DANGER" in letters not less than 3.2 mm (1/8 inch) high. The remaining letters shall not be less than 1.6 mm (1/16 inch) high.
3. Permanent methods include Molded, Die-stamped, Paint-stenciled, Stamped, etched metal or pressure-sensitive label recognized by NRTL.

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ILLUSTRATIONS

Both English and French instructions shall be provided. Instruction shall include below information:

1. Proper installation method.
2. Wiring instructions that specify the proper method of connecting the grounding means and maintaining polarity shall be included.
3. Other information should not lead to misuse.
4. Electrical Ratings and Descriptions.

According to Table 65.1 of UL 1741 (Third Edition, Dated September 28, 2021)		
Model:	R8KLNA-P4	R12KLNA-P4
INPUT RATINGS:		
Maximum input voltage	500 V dc	
Range of input operating voltage (without Battery)	120 V dc to 500 V dc	120 V dc to 500 V dc
Range of input operating voltage (with Battery)	120V dc to 430 V dc	120V dc to 430 V dc
Maximum input current (dc)	16A/16A/16A/16A	16A/16A/16A/16A
Number of input	4	
OUTPUT RATINGS (Grid terminal, Bi-directional):		
Output power factor rating	default >0.99 (-0.8~0.8 adjustable)	
Operating voltage range (ac) (L-L)	0.88Un~1.1Un	
Operating frequency range or single frequency	58.5 Hz to 61.2 Hz	
Number of phases	Single phase/ Split phase	
Nominal output voltage (ac)	120Vac/240Vac, split phase, 240V single phase	
Normal output frequency	60 Hz	
Maximum continuous output current (ac) per line	36.7 Arms	55.0 Arms
Rated output current (ac) per line	33.3 Arms	55.0 Arms
Maximum continuous output power (ac)	8.8 kVA	13.2 kVA
Rated continuous output power (ac)	8.0 kW	12.0 kW
Maximum output fault current (ac) and duration	378Apk, 83ms, 71.1 Arms@1cycle; 97.7 Arms@3cycle; 97.7 Arms@5cycle	
Trip limit and trip time accuracy - Voltage:	±2.4 V	±2.4 V
Utility interconnection voltage and frequency trip limits and trip times	see Note 1 and 2	
Trip limit and trip time accuracy - Frequency:	±0.01 Hz	
Trip limit and trip time accuracy - Time	±1%setting, but not less than 50ms	
Normal operation temperature range	-25°C to 60°C	
Enclosure Rating Type	Type 3R	
Weigh (kg)	55kg	55kg

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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 57 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Revised 03-September-2021

According to Table 65.1 of UL 1741 (Third Edition, Dated September 28, 2021)		
Model:	R8KLNA-P4	R12KLNA-P4
Dimension (mm)	495*900*260	
OUTPUT RATINGS (BACKUP output terminal):		
Number of phases	Single phase/Split phase	
Nominal output voltage (ac)	120Vac/240Vac	
Normal output frequency	60 Hz	
Maximum continuous output power (ac)	8.8 kVA	13.2kVA
Rated output power (ac)	8.0 kW	12.0 kW
Battery terminal, Bi-directional:		
Battery Type	Lithium/Lead acid	
Range of DC operating voltage (Vdc)	40-58V dc	
Nominal voltage (Vdc)	48V dc	
Max. charging/ discharging current (Adc)	170/170 Arms	250/260 Arms
Max. charging/ discharging power (W)	8000W	12000W

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Model:

R16KLNA-P4

R8KLNA-P6

INPUT RATINGS:		
Maximum input voltage	500 V dc	
Range of input operating voltage (without Battery)	120 V dc to 500 V dc	120 V dc to 500 V dc
Range of input operating voltage (with Battery)	120V dc to 430 V dc	120V dc to 430 V dc
Maximum input current (dc)	20A/20A/20A/20A	16A/16A/16A/16A
Number of input	4	
OUTPUT RATINGS:		
Output power factor rating	default >0.99 (-0.8~0.8 adjustable)	
Operating voltage range (ac) (L-L)	0.88Un~1.1Un	
Operating frequency range or single frequency	58.5 Hz to 61.2 Hz	
Number of phases	Single phase/ Split phase	
Nominal output voltage (ac)	120Vac/240Vac	
Normal output frequency	60 Hz	
Maximum continuous output current (ac) per line	73.3 Arms	36.7 Arms
Rated output current (ac) per line	66.7 Arms	33.3 Arms
Maximum continuous output power (ac)	17.6 kVA	8.8 kVA
Rated continuous output power (ac)	16.0 kW	8.0 kW
Maximum output fault current (ac) and duration	378Apk, 83ms, 71.1 Arms@1cycle; 97.7 Arms@3cycle; 97.7 Arms@5cycle	
Trip limit and trip time accuracy - Voltage:	±2.4 V	±2.4 V
Utility interconnection voltage and frequency trip limits and trip times	see Note 1 and 2	
Trip limit and trip time accuracy - Frequency:	±0.01 Hz	
Trip limit and trip time accuracy - Time	±1%setting, but not less than 50ms	
Normal operation temperature range	-25°C to 60°C	
Enclosure Rating Type	Type 3R	
Weigh (kg)	55kg	56kg
Dimension (mm)	495*900*260	
OUTPUT RATINGS (BACKUP output terminal):		
Number of phases	Single phase/Split phase	
Nominal output voltage (ac)	120Vac/240Vac	
Normal output frequency	60 Hz	
Maximum continuous output power (ac)	13.2 kVA	8.8 kVA
Rated output power (ac)	13.0 kW	8.0 kW
Battery terminal, Bi-directional:		
Battery Type	Lithium/Lead acid	
Range of DC operating voltage (Vdc)	40-58V dc	
Nominal voltage (Vdc)	48V dc	
Max. charging/ discharging current (Adc)	260/280 Arms	170/170 Arms
Max. charging/ discharging power (W)	16000W	8000W

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Revised on: September 2017

Model:	R12KLNA-P6	R16KLNA-P6
INPUT RATINGS:		
Maximum input voltage	500 V dc	
Range of input operating voltage (without Battery)	120 V dc to 500 V dc	120 V dc to 500 V dc
Range of input operating voltage (with Battery)	120V dc to 430 V dc	120V dc to 430 V dc
Maximum input current (dc)	16A/16A/16A/16A	20A/20A/20A/20A
Number of input	4	
OUTPUT RATINGS:		
Output power factor rating	default >0.99 (-0.8~0.8 adjustable)	
Operating voltage range (ac) (L-L)	0.88Un~1.1Un	
Operating frequency range or single frequency	58.5 Hz to 61.2 Hz	
Number of phases	Single phase/ Split phase	
Nominal output voltage (ac)	120Vac/240Vac	
Normal output frequency	60 Hz	
Maximum continuous output current (ac) per line	55.0 Arms	73.3 Arms
Rated output current (ac) per line	55.0 Arms	66.7 Arms
Maximum continuous output power (ac)	13.2 kVA	17.6 kVA
Rated continuous output power (ac)	12.0 kW	16.0 kW
Maximum output fault current (ac) and duration	378Apk, 83ms, 71.1 Arms@1cycle; 97.7 Arms@3cycle; 97.7 Arms@5cycle	
Trip limit and trip time accuracy - Voltage:	±2.4 V	±2.4 V
Utility interconnection voltage and frequency trip limits and trip times	see Note 1 and 2	
Trip limit and trip time accuracy - Frequency:	±0.01 Hz	
Trip limit and trip time accuracy - Time	±1%setting, but not less than 50ms	
Normal operation temperature range	-25°C to 60°C	
Enclosure Rating Type	Type 3R	
Weigh (kg)	56kg	56kg
Dimension (mm)	495*900*260	
OUTPUT RATINGS (BACKUP output terminal):		
Number of phases	Single phase/Split phase	
Nominal output voltage (ac)	120Vac/240Vac	
Normal output frequency	60 Hz	
Maximum continuous output power (ac)	13.2kVA	13.2 kVA
Rated output power (ac)	12.0 kW	13.0 kW
Battery terminal, Bi-directional:		
Battery Type	Lithium/Lead acid	
Range of DC operating voltage (Vdc)	40-58V dc	
Nominal voltage (Vdc)	48V dc	
Max. charging/ discharging current (Adc)	250/260 Arms	260/280 Arms
Max. charging/ discharging power (W)	12000W	16000W

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According to Table 28 of IEEE 1547-2018 (representative)			
#	Parameter	EUT Configuration Information	
1	Active power rating at unity power factor (nameplate active power rating)	12.0kW	
2	Active power rating at specified over-excited power factor	10.56 kW	
3	Specified over-excited power factor	0.8	
4	Active power rating at specified under-excited power factor	10.56 kW	
5	Specified under-excited power factor	0	
6	Apparent power maximum rating	13.2 kVA	
7	Normal operating performance category	B	
8	Abnormal operating performance category	III	
9	Reactive power injected maximum rating	7920 VAr	
10	Reactive power absorbed maximum rating	7920 VAr	
11	Active power charge maximum rating	12.0 kW	
12	Apparent power charge maximum rating	13.2 kVAr	
13	AC voltage nominal rating	240 V	
14	AC voltage maximum rating	264 V	
15	AC voltage minimum rating	211.2 V	
16	Supported control mode functions	Item 0~26	
17	Reactive susceptance that remains connected to the Area EPS in the cease to energize and trip state	0	
18	Manufacturer	59923	
19	Model	R12KLNA-P6	
20	Serial number	M10824400001	
21	Version	Ver 1.0406(DSP), Ver 2.0306(ARM)	
According to Table 3 of IEEE 1547-2018			
Time Frame		Steady-state measurements	
Parameter	Minimum measurement accuracy	Measurement window	Range
Voltage, RMS	± 1% Vnom	10 cycles	0.5 p.u. to 1.2 p.u.
Frequency	0.01 Hz	60 cycles	50 Hz to 66 Hz
Active Power	± 5% Srated	10 cycles	-1.0 p.u. to 1.0 p.u.
Reactive Power	± 5% Srated	10 cycles	-1.0 p.u. to 1.0 p.u.
Time	1). 1% of measured duration (>= 5 s) 2). 30 ms (< 5 s)	N/A	0~ 1000 s

Note: Maximum reactive power level (injection and absob) is 60%Sn for function Voltage-reactive power mode (Q(U)) and Active power-reactive power mode (P-Q), while 100%Sn for Constant power factor mode (PF) and Constant reactive power mode.

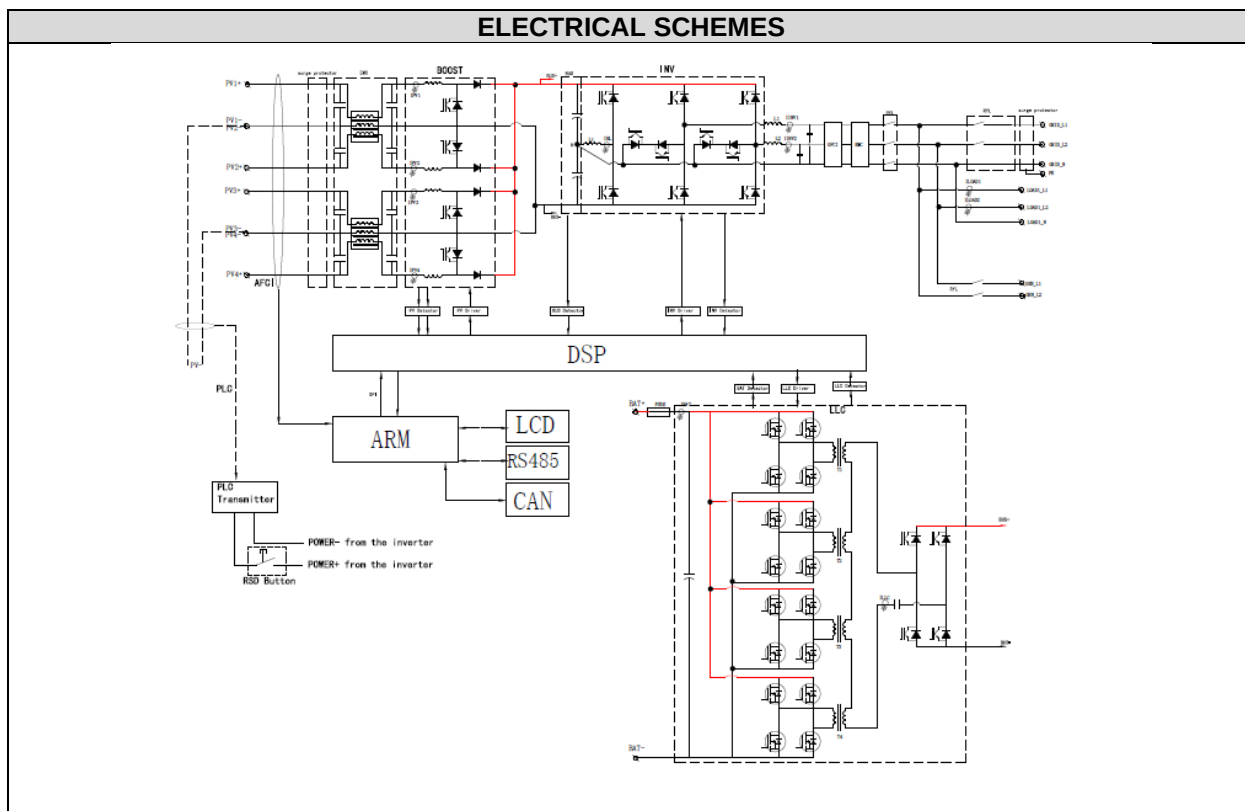
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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 61 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

Note 1: Utility Interconnection Voltage and Frequency Trip Limits and Trip Times:

According to IEEE 1547-2018				
Shall trip—Category III				
Shall trip function	Default settings		Ranges of allowable settings	
	Voltage (p.u.)	Clearing time (s)	Voltage (p.u.)	Clearing time (s)
OV2	1.20	0.16	fixed at 1.20	fixed at 0.16
OV1	1.10	13.0	1.10 ~ 1.20	1.0 ~ 13.0
UV1	0.88	21.0	0.0 ~ 0.88	2.0 ~ 50.0
UV2	0.50	2.0	0.0~0.50	0.16 ~ 21.0
OF2	62.0	0.16	61.8~66.0	0.16~1000.0
OF1	61.2	300.0	61.0~66.0	180.0~1000.0
UF1	58.5	300.0	50.0~59.0	180.0~1000.0
UF2	56.5	0.16	50.0~57.0	0.16~1000.0



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SUMMARY OF TEST RESULTS

The following tests were performed:

UL 1741 (Third Edition, Dated September 28, 2021)

Section	Test Description
26	Spacings
27	Alternate Spacings – Clearances and Creepage Distances
45	Maximum - Voltage Measurements
46	Temperature
47	Dielectric Voltage-Withstand Test
48.2	Output Power Characteristics – Output ratings
48.3	Output Power Characteristics – DC input range
48.4	Harmonic distortion
50.2	Abnormal Tests – Output overload test
50.3	Abnormal Tests – Short Circuit test
50.4	Abnormal Tests – DC input miswiring test
50.5	Abnormal Tests – Ventilation test
50.6	Abnormal Tests – Component Short and open circuit
50.8	Loss of control circuit
51	Grounding Impedance test
60	Capacitor Voltage Determination Test
61	Stability
63	Compression test
64	Rain and Sprinkler Tests
76	Normal operations
79	Connection sequence
80.2	Input and output faults
80.3	Charge controller mis-wiring
80.4	Low-voltage disconnect

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UL 1741 (Third Edition, Dated September 28, 2021)

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51	Grounding Impedance test
60	Capacitor Voltage Determination Test
61	Stability
63	Compression test
64	Rain and Sprinkler Tests
98.1	Operational tests for PVRSS/PVRSE verification of levels – controlled conductors
98.2	Verification testing of PVRSS at rated extremes
98.3	Power supply grid support ride through
98.4	Inverters rated as PVRSE

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SUMMARY OF TEST RESULTS(Continuing)

CSA C22.2 No. 107.1:16 (reaffirmed 2021)

Section	Test Description
6.2	Ratings
6.3	Temperature
6.5	Dielectric strength
6.6	Abnormal operation
6.9	Compression
10.5.1	AC output voltage for stand-alone inverters
10.5.2	Harmonic distortion for stand-alone inverters
10.5.3	DC injection
10.5.4	Frequency
10.5.5	Stand-alone inverter dc output short-circuit current contribution
14.3.2	Output ratings
14.3.3	Harmonic current distortion
14.3.4	Utility voltage and frequency variation test
14.3.5	Anti-islanding test
14.3.6	Loss of control circuit power
14.3.7	Maximum back feed current into the PCE input circuit
14.3.8	Testing of automatic disconnecting mean for non-isolated inverters
14.3.9	AC output short circuit current contribution test

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 65 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

SUMMARY OF TEST RESULTS(Continuing)

UL 1741SB (Third Edition, Dated September 28, 2021) Based upon IEEE 1547-2018 and IEEE 1547.1-2020			
UL1741SB (Ed.3)	IEEE1547: 2018	IEEE Std 1547.1:2020	
SB4.3.5.3	4.4	5.3	Temperature stability
SB4.3.5.4	6.4	5.4	Test for response to voltage disturbances
SB4.3.5.4.2	6.4.1	5.4.2	Test for overvoltage trip
SB4.3.5.4.3		5.4.3	Test for undervoltage trip
SB4.3.5.4.4	6.4.2.1	5.4.4	Low-voltage ride-through tests
	6.4.2.3		
	6.4.2.2	5.4.5	Test for voltage disturbances within continuous operating region
SB4.3.5.4.7	6.4.2.1	5.4.7	High-voltage ride-through tests
	6.4.2.4		
SB4.3.5.5	6.5	5.5	Test for response to frequency disturbances
SB4.3.5.5.1	6.5.1	5.5.1	Test for overfrequency trip
SB4.3.5.5.2		5.5.2	Test for underfrequency trip
SB4.3.5.5.3	6.5.2.1	5.5.3	Test for low-frequency ride-through
	6.5.2.3		
SB4.3.5.5.4	6.5.2.1	5.5.4	Test for high-frequency ride-through
	6.5.2.4		
	6.5.2.5	5.5.5	Test for rate of change of frequency (ROCOF)
SB4.3.5.5.6	6.5.2.6	5.5.6	Test for voltage phase-angle change ride-through
	4.10.2	5.6	Enter service
	4.10.3		
	6.6		
	4.10.4	5.7.4	Synchronization control function test for equipment with no synchronizing disable capability (variation 3)
	4.11	5.8	Interconnection integrity
SB4.3.5.8.1	4.11.1	5.8.1	Protection from electromagnetic interference (EMI) test
SB4.3.5.8.2	4.11.2	5.8.2	Surge withstand performance test
	4.11.3	5.8.3	Paralleling device tests
	7.1	5.9	Limitation of dc injection for inverters
SB4.3.5.10	8.1	5.10	Unintentional islanding
SB4.3.5.10.2	8.1.1	5.10.2	Balanced generation to load unintentional islanding test
	6.2.2	5.11	Open phase
	7.3	5.12	Current distortion
		5.12.2	Current distortion test for DER not capable of operation isolated from external source
	4.6.2	5.13	Limit active power
	5.3	5.14	Voltage regulation
SB4.3.5.14.3	5.3.2	5.14.3	Test for constant power factor (p.f.) mode
SB4.3.5.14.4	5.3.3	5.14.4	Test for voltage-reactive power (volt-var) mode
SB4.3.5.14.5		5.14.5	Test for voltage-reactive power (volt-var) mode (VRef test)

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 66 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

UL 1741SB (Third Edition, Dated September 28, 2021) Based upon IEEE 1547-2018 and IEEE 1547.1-2020			
UL1741SB (Ed.3)	IEEE1547: 2018	IEEE Std 1547.1:2020	
SB4.3.5.14.6		5.14.6	Test for voltage - reactive power (volt-var) mode with an imbalanced grid
SB4.3.5.14.7	5.3.4	5.14.7	Test for active power-reactive power mode (watt-var)
SB4.3.5.14.8	5.3.5	5.14.8	Test for constant reactive power (var) mode
SB4.3.5.14.9	5.4.2	5.14.9	Test for voltage-active power (volt-watt) mode
		5.14.10	Test for voltage-active power (volt-watt) mode with an imbalanced grid
SB4.3.5.15		5.15	Frequency support
SB4.3.5.15.2	6.5.2.7	5.15.2	Test for frequency-droop (frequency-power or frequency-watt) capability—above nominal frequency
SB4.3.5.15.3		5.15.3	Test for frequency-droop (frequency-power or frequency-watt) capability—below nominal frequency
SB4.3.5.16	4.7	5.16	Test for prioritization of DER responses
SB4.3.5.17	7.4	5.16.2	Test for voltage and frequency regulation priority
SB4.3.5.17.2	7.4.2	5.17	Limitation of overvoltage contribution
	11.4	5.17.2	Load rejection overvoltage (LROV) test
	11.4.2	5.18	Fault current tests
		5.18.1	Fault current tests for inverters
SB4.3.6	10 ⁽⁴⁾	5.19	Persistence of DER parameter settings
SB4.3.6.8.2		6	Interoperability tests (IEEE2030.5 Protocol used)
		6.8.2	
		6.4	Nameplate data test
		6.8.2.1	
SB4.3.6.6		6.6	Monitoring information test
		6.8.2.3	
SB4.3.6.7		6.7	Management information test
		6.8.2.4	
UL1741SB (Ed.3)	IEEE1547a: 2020	IEEE Std 1547.1:2020	
	6.4.1	5.4.3	Test for undervoltage trip
UL1741SB (Ed.3)	SRD 2.0	IEEE Std 1547.1:2020	
	PART II.A	5.6	Enter service
	PART II.A	5.5.3	Test for low-frequency ride-through
		5.5.4	Test for high-frequency ride-through
	PART II.C	5.15.2	Test for frequency-droop (frequency-power or frequency-watt) capability - abovenominal frequency
		5.15.3	Test for frequency-droop (frequency-power or frequency-watt) capability - below nominal frequency

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 67 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

IEEE 2030.5 Protocol Gateway Information:

Communication Protocol
Product Type
Product Name
Product Model
Product Object ID
Software Operation Environment Type
Software Name
Software Version
Software Checksum
Operating System
Operating System Version
SunSpec Certificate Number

IEEE 2030.5 (SEP2)
CA Rule 21/CSIP DER Client
Stick Logger(WiFi)
LSW-5 Series
1.3.6.1.4.1.58214.01
Physical device
LSW5BLE_MW17_CSIP5406_1.11
1.11
16b8d9d19a489cfd35277816eae40232
FreeRTOS
V10.2.0
CS-000046

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SUMMARY OF TEST RESULTS(Continuing)

UL 1699B: First Edition, Dated August 22, 2018

Section	Test Description
25	HUMIDITY CONDITIONING
27	Voltage Surge Test
27.2	UNWANTED TRIPPING TEST (RING WAVE)
27.3	UNWANTED TRIPPING TEST (RING WAVE)
28	Environmental Test Sequence
29	Arc Fault Detection Tests
30	Unwanted Tripping Tests
30.2	LOADING CONDITION I – INVERTERS, CONVERTERS, AND CHARGE CONTROLLERS
30.3	Loading condition II – DC switch operation
30.4	Loading condition III – Irradiance step changes
31	NORMAL TEMPERATURE TEST
35	Dielectric Voltage-Withstand Test
42	RESISTANCE TO ENVIRONMENTAL NOISE TEST
56	OVERVOLTAGE, OVERLOAD, ENDURANCE, ABNORMAL OPERATIONS, AND SHORT CIRCUIT TESTS

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 69 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

SUMMARY OF TEST RESULTS(Continuing)

**UL 1741SA (Third Edition, Dated September 28, 2021) setting according to CPUC
Rule 21 Effective on Jul 30, 2017**

Section	Test Description
SA8	Anti-islanding Protection - Unintentional Islanding with Grid Support Functions Enabled
SA9	L/HVRT Low and High Voltage Ride-Through
SA10	L/HFRT Low and High Frequency Ride-Through
SA11	RR – Normal Ramp Rate and SS – Soft-Start Ramp Rate
SA12	SPF – Specified Power Factor
SA13	Volt/VAr Mode (Q(V))
SA14	Frequency-Watt (FW)
SA15	Volt-Watt (VW)
SA17	Limit Active Power
SA18	Disable Permit Service

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 70 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

SUMMARY OF TEST RESULTS(Continuing)

CSA C22.2 No. 330:23 (2023) Photovoltaic rapid shutdown systems

Section	Test Description
5.2	Testing PVRSSs or PVRSE — Voltage levels

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 71 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

SUMMARY OF TEST RESULTS(Continuing)

UL 1741 (April 20, 2010 Ed.2) CRD, Subject: Power Control Systems (PCS)

Section	Test Description
203.5	Step Change in Load Test
203.6	Step Change in Generation Test
204	Optional Test
205	Abnormal Tests

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REVISION TABLE

The following changes have been made to this Report:

<u>Revision No.</u>	<u>Date</u>	<u>Project #</u>	<u>Revision prepared by</u>	<u>Page</u>	<u>Description of Change</u>
00	2024-05-10	GZES2401001726PV	Harry Zou	All	First issuance.
01	2024-08-28	GZES2407013689PV	Harry Zou	All	Changed the revised date from None to 2024-08-28.
				2	Changed factory information from Shenzhen INVT Electric Co., Ltd. Bao'an Branch Factory to Shenzhen Megarevo Technology Co., Ltd.
				4-5	Changed the batteries side charge/discharge current of model 16KLNA-P4 from "250A/260Amax" to "260A/280Amax", model 16KLNA-P6 from "250A/260Amax" to "260A/280Amax".
				51-53	Updated nameplate labels for models R8KLNA-P4, R12KLNA-P4, R16KLNA-P4, R8KLNA-P6, R12KLNA-P6, R16KLNA-P6
				56-59	Updated electrical Ratings table
02	2024-09-05	GZES2401001727PV	Harry Zou	All	Modified Revised date from 2024-08-28 to 2024-09-05.
				1	Added standard: 1. Photovoltaic (PV) DC Arc-Fault Circuit Protection-UL 1699B: First Edition, Dated August 22, 2018 2. UL 1741 Supplement SA - Grid Support Utility Interactive Inverters and Converters (2010 Ed.2 Rev Feb. 15, 2018) 3. CSA C22.2 NO. 330-23 (2023) 4. Generating Facility Interconnections- CPUC Rule 21 Effective on Jul 30, 2017 5. UL1741 SECTION 92 TO SECTION 99, SEPTEMBER 28, 2021 - PHOTOVOLTAIC RAPID SHUTDOWN SYSTEMS
				5	Changed date(s) of performance of tests

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LISTING REPORT/INSPECTION PROCEDURE

SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 73 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

				68	Added SUMMARY OF TEST RESULTS of Photovoltaic (PV) DC Arc-Fault Circuit Protection-UL 1699B: First Edition, Dated August 22, 2018
				69	Added SUMMARY OF TEST RESULTS of UL 1741 Supplement SA - Grid Support Utility Interactive Inverters and Converters (2010 Ed.2 Rev Feb. 15, 2018)
				70	Added SUMMARY OF TEST RESULTS of CSA C22.2 NO. 330-23 (2023)
				73	Added standard: 1. Photovoltaic (PV) DC Arc-Fault Circuit Protection-UL 1699B: First Edition, Dated August 22, 2018 2. UL 1741 Supplement SA - Grid Support Utility Interactive Inverters and Converters (2010 Ed.2 Rev Feb. 15, 2018) 3. CSA C22.2 NO. 330-23 (2023) 4. Generating Facility Interconnections- CPUC Rule 21 Effective on Jul 30, 2017 5. UL1741 SECTION 92 TO SECTION 99, SEPTEMBER 28, 2021 - PHOTOVOLTAIC RAPID SHUTDOWN SYSTEMS
03	2024-09-26	GZES2401001731PV	Harry Zou	All	Modified Revised date from 2024-09-05 to 2024-09-26.
				1	Added standard: UL 1741 (April 20, 2010 Ed.2) CRD, Subject: Power Control Systems (PCS)
				71	Added SUMMARY OF TEST RESULTS of UL 1741 (April 20, 2010 Ed.2) CRD, Subject: Power Control Systems (PCS)
				74	Added standard: UL 1741 (April 20, 2010 Ed.2) CRD, Subject: Power Control Systems (PCS)

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SHENZHEN MEGAREVO TECHNOLOGY CO., LTD
PROJECT NUMBER: GZ-CERT240200408

Page 74 of 74
Issued: May 10, 2024
Revised 03: September 26, 2024

CONCLUSION

Samples of the products covered by this Report have been found to comply with the applicable requirements of Standard:

- Inverters, Converters, Controllers and Interconnection System Equipment for Use With Distributed Energy Resources –UL 1741. (Third Edition, Dated September 28, 2021)
- Power conversion equipment – CSA C22.2 No. 107.1 issued on June 2016
- UL 1741 Supplement SB - Grid Support Utility Interactive Inverters and Converters Based upon IEEE 1547-2018 and IEEE 1547.1-2020. (Third Edition, Dated September 28, 2021)
- Photovoltaic (PV) DC Arc-Fault Circuit Protection-UL 1699B: First Edition, Dated August 22, 2018
- UL 1741 Supplement SA - Grid Support Utility Interactive Inverters and Converters (2010 Ed.2 Rev Feb. 15, 2018)
- CSA C22.2 NO. 330-23 (2023)

Reference standard

- IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces – IEEE Std 1547:2018 (Revision of IEEE Std 1547:2003), approved 15 February 2018; IEEE Std 1547a™-2020 Approved 9 March 2020. (Refer to IEEE Std 1547.1:2020, approved 5 March 2020.)
- Hawaiian Electric Companies, IEEE 1547.1-2020 Source Requirements Document Version 2.0 ("SRD V2.0"), effective on July 1st, 2020
- Generating Facility Interconnections- CPUC Rule 21 Effective on Jul 30, 2017
- UL1741 SECTION 92 TO SECTION 99, SEPTEMBER 28, 2021 - PHOTOVOLTAIC RAPID SHUTDOWN SYSTEMS
- UL 1741 (April 20, 2010 Ed.2) CRD, Subject: Power Control Systems (PCS)

Report Prepared by:



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Roger Hu
Global Reviewer

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